

Alternating Baxter involutions (OEIS A347546)

The corrected count, a correction of Min’s 2021 recurrence,
asymptotics and inverses, and the refinement by pairs of fixed points

Two research reports of one external research session
(bundle Reports 149 and 151), merged into one ProveIt report

Parts I and II: 3 October 2026 merged: 5 October 2026

Abstract

A permutation is doubly alternating if it and its inverse both begin with a rise and alternate, $\pi_1 < \pi_2 > \pi_3 < \dots$; OEIS A347546 is named “Number of involutions of doubly alternating Baxter permutations of length n ”. This report prints two manuscripts as Parts I and II. Part I shows that the recurrence of Min’s paper (J. Chungcheong Math. Soc. 34 (2021), Theorem 2.7), from which the data of A347546 were computed, miscounts this class: in an even-length involution the two outer blocks are inverses of each other, not involutions, so the count of outer choices is the Catalan number C_j , not the number of involutions o_j , and $C_4 = 14 > 12 = o_4$. The class has 2168 members of length 20, not 2166; two explicit omitted involutions are given. From three published structural lemmas of Min and Park (2006) Part I proves the corrected recurrences, which re-derive along the way Guibert and Linusson’s Catalan count of doubly alternating Baxter permutations; the generating functions are algebraic with dominant singularity $\rho = (\sqrt{5} - 1)/4$, the even and odd subsequences have equivalents $\mathcal{A}_s(1 + \sqrt{5})^m m^{-3/2}$ with explicit constants and expansions to every fixed order, and their thresholds are inverted by Lambert W_{-1} with ceiling brackets that leave at most two adjacent candidates. Part II refines the count by pairs of fixed points ($q = 1$ gives Part I back): positive coefficient factorizations, expansions to every fixed order uniform on compact complex windows of the sparse weight $q = w/m$ with entire profiles, an exact criterion for inverting in the weight, interior inverse expansions, and Poisson-type limit laws. The write adds a proof that the recurrence-defined sequence lies strictly below the class count at every length $n = 20$ and $n \geq 22$, so that 21 of the 42 terms of A347546 are wrong (its definition is right); nothing has been submitted to the OEIS. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 47fc7a069 (batch 106). They study one class, the alternating Baxter involutions:

Part	Bundle report	Delivered title	Labels
I	149 (base)	Corrected enumeration and asymptotics of alternating Baxter involutions: the class intended by OEIS A347546 and its parity inverses	abx:en:
II	151	Sparse fixed point crossover for alternating Baxter involutions: entire profiles, exact thresholds, and weighted inverses	abx:fp:

Arrangement. Report 149 is the base: it alone proves the structural decomposition, the correction and the length asymptotics, and its **Report149.tex** was staged as this **article.tex**. Report 151 refines its enumeration by a second variable and imports its decomposition (“those established in Report 149”), so it is printed second, as Part II. The two Parts prove different theorems; neither supersedes the other.

Embedded copy. Report 151’s archive carries Report 149’s source and PDF under **foundation/**; both files are byte-identical to the standalone delivery (checked at placement and again at the write). They are printed once, as Part I, and are not shipped.

Numbering. Sections are numbered continuously, and statements and equations within sections, as in both manuscripts. Part I keeps Report 149’s Sections 1–11 and every number in them. Report 151’s Section k is Section $k + 11$ here (Sections 12–22), and its statement or equation $k.j$ is $(k + 11).j$: so its Theorem 2.1 is Theorem 13.1, its Theorem 3.1 is Theorem 14.1, its Proposition 7.2 is Proposition 18.2, and its equation (2.1) is (13.1). Section 23 was added in the write; no delivered number moved.

Notes added in the write are set in small type and tagged [**write**], with their date (5 October 2026). They give cross-references between the Parts, credit prior work, mark what Part II restates from Part I, and record what the write checked. No delivered statement was changed and no delivered symbol was renamed; Report 151’s opening restatement of Report 149 is printed once, in Part I (“Provenance and merge decisions” below). Neither manuscript has a section of open questions; Section 23 records the corrections of the published literature with their proofs, and gathers, under Vladimir’s standing rule of 4 October 2026, every claim the two manuscripts leave unproved, with its source, sketch and what is missing.

What is proved, and where

Here a_n is the number of involutions in the class $\mathcal{B}_n$ of doubly alternating Baxter permutations of length n , $e_m = a_{2m}$, $o_m = a_{2m+1}$, r_m the number of involutions in the doubly reverse-alternating class $\mathcal{R}_{2m}$ (so $o_m = r_m$), C_j the Catalan numbers, and $e_m(q)$, $r_m(q)$ Part II’s refinements by pairs of fixed points.

Statement	Where
$ \mathcal{B}_{2j} = \mathcal{R}_{2j} = C_j$ (published: Guibert–Linusson 2000, Min–Park 2006)	re-derived in Part I, Section 3, (3.3), from Min–Park’s necessity lemmas
Corrected recurrences $o_m = \sum_i e_i o_{m-1-i}$, $e_m = o_{m-1} + \sum_j C_j e_{m-2j-2}$; $O = 1/(1 - xE)$, $E = 1 + xO + x^2 EC(x^2)$	Part I, Theorem 1.1 (Sections 2–4), assuming three statements of Min–Park
$a_{20} = 2168$, not 2166; two explicit omitted involutions; the deficit is exactly 2	Part I, Section 5
Min’s Theorem 2.7: odd formula true, even formula false exactly for $n \geq 10$; the recurrence sequence is below the class at $n = 20$ and every $n \geq 22$, equal otherwise	[write] Remark 23.1, Proposition 23.2
Radicals for E , O ; quartic relation; dominant singularity $\rho = (\sqrt{5} - 1)/4$, Delta domain	Part I, Section 6
$s_m = \mathcal{A}_s \Lambda^m m^{-3/2} (1 + \sum_j c_{s,j} m^{-j} + \dots)$, $\Lambda = 1 + \sqrt{5}$, every fixed order, $\mathcal{A}_s$, λ_s exact; ratios; eventual monotonicity	Part I, Theorem 7.1, Section 7
Length inverses: Lambert W_{-1} centre, every fixed inverse order; safe ceilings; at most two adjacent candidates	Part I, Theorems 8.1, 9.1, Corollary 9.2
Bivariate system $E = (1 + xqR)/M$, $R = 1/(1 - xE)$ ($q = 1$: Theorem 1.1); positive factorizations; $[q]e_m = 2^{m-1}h_m$; degrees	Part II, Theorem 13.1, Subsection 13.1
Every fixed order of $T_{s,m}(w)$, uniform on compact complex w -discs; entire profiles and their finite algorithms	Part II, Theorem 14.1, Sections 15–17
Exact nonnegative inverse in the weight; the odd- m obstruction; two boundary roots	Part II, Theorem 18.1, (18.2), Proposition 18.2
Interior inverse in powers of $m^{-1/2}$; leading inverses (one with W_{-1})	Part II, Theorem 19.1, (19.5), (19.6)
Sparse Poisson-type laws, total variation $O(m^{-1/2})$	Part II, Theorem 20.1
<i>Open in both Parts</i>	
Effective constants and onsets (certified brackets); all-index monotonicity of a_n ; the nature of the recurrence-defined sequence; Min’s 2002 thesis (priority); a proof of Min–Park’s necessity lemmas inside the report; a central limit theorem at fixed weight, growing windows, boundary-uniform and global inverses; uniformity in the order and convergence. See Section 23.	

The companions’ finite computations (exact integer and rational arithmetic, finite enumerations, high-precision diagnostics) check identities, coefficients and the two omitted involutions; no all-length or asymptotic statement rests on them.

The published recurrence, the OEIS entry, and the inverses

Min (2021). S. Min, *The enumeration of involutions of doubly alternating Baxter permutations*, J. Chungcheong Math. Soc. **34**(3) (2021) 253–257 [2], writes B_n for $\mathcal{B}_n$, RB_n for $\mathcal{R}_n$, $B_n(I)$ for the involutions in B_n , and $b_n = |B_n(I)|$, which is a_n here. Its Theorem 2.7, its proof and its list of values are quoted and corrected in Section 23: the odd formula is right; the even formula, and with it every value from b_{20} on except b_{21} , is wrong. The write read the paper (the publisher’s PDF, whose SHA-256 equals the one recorded in `data/149-enum-SOURCE_PROVENANCE.json`).

The OEIS entry. A347546 as read by the write on 5 October 2026 (revision #20 of June 29, 2022): name “Number of involutions of doubly alternating Baxter permutations of length n.”, author line *Sook Min, Sep 06 2021*, offset 0, data for $n = 0, \dots, 41$ (the b-file is “synthesized from sequence

entry”), links to Barnabei et al. and to Min (2021), a Python program, and “Cf. A001181” (the Baxter numbers). The data and the program implement Min’s recurrence: the program reproduces the 42 data terms (checked by the write), and Proposition 23.2 shows that they differ from the class count exactly at $n = 20$ and $n = 22, \dots, 41$. The *name* describes the class this report counts. Remark 23.5 tabulates the 21 wrong terms. OEIS data are available under CC BY-SA 4.0; this report records the correction and submits nothing to the OEIS.

Barnabei et al. Barnabei, Bonetti, Castronuovo and Silimbani [5] (arXiv:2206.13877; ECA 3:1 (2023) S2R4) identify, in their Section 10, the alternating involutions avoiding 2413 and 3142 with the alternating Baxter involutions (through Ouchterlony’s theorem) and write: “A recurrence relation for this sequence has been found in [13] and the corresponding sequence in [17] is A347546” ([13] is Min (2021), [17] the OEIS). They print no value of the sequence. Their recalled Baxter definition uses weak inequalities between the four indices, which Part I does not adopt (Section 1).

The inverses. Part I inverts in the *length*: by (8.2), $\log s_m = bm - p \log m + \log \mathcal{A}_s + \dots$ with $b = \log \Lambda$ and $p = 3/2$. Its centre X_s of (8.3) is the large solution of $aX + b' \log X = L$ with $a = \log \Lambda$, $b' = -3/2$ and $L = \log(y/\mathcal{A}_s)$: an instance of the Lambert core `p0:thm:lambert-core` (3) of the transseries volume [13], branch $b' < 0$, which selects W_{-1} exactly as Part I does. The formal correction (8.4) is that volume’s equation `p0:eq:formal-exactness` with $q_j = d_{s,j}$, so the coefficients $v_{s,k}$ of (8.5) are the coefficients c_n of `p0:thm:lambert-centered`; and Theorem 8.1 is the analytic part (5) of that theorem applied to the explicit model $\exp F_{s,J}(x)$, which is of exponential–power type with data $(\mathcal{A}_s, \log \Lambda, 1, -3/2, 1, \sum_{j \leq J} d_{s,j} t^j)$ (instance, for $K \leq J$). The ceiling brackets of Theorem 9.1 follow the separation pattern of `p0:thm:staircase` (2), and the identity (9.5) has the form of its residue-class part (4) with $r = 2$; but Part I proves both directly at the integers, choosing no interpolation of e_m or o_m , so they are *not* instances of that theorem, which presupposes one. Corollary 9.2 has no counterpart in the volume. Part II inverts in the *weight* w at fixed half-length m : its exact criterion (Theorem 18.1) is a statement about polynomials with nonnegative coefficients; its interior and boundary expansions (Theorem 19.1, Proposition 18.2) are reversions of a root with nonzero derivative in powers of $m^{-1/2}$, analogous to the volume’s perturbed inversion `p0:thm:perturbed-inversion` but not instances of it (the perturbation here is an asymptotic expansion, not an analytic family); its ascent leading inverse (19.5) is a logarithm; and its reverse leading inverse (19.6) is an instance of `p0:thm:lambert-core` (3) after the change of variable $X = w_0 + 1/h$, with $a = 1$, $b = -1$, $L = \log h + 1/h$ (proof in the note after (19.6)). Neither manuscript cites the volume or claims novelty for the inversion.

Notation across the two Parts

Common to both Parts: $C(t) = \sum C_j t^j$ and $M(x) = 1 - x^2 C(x^2)$ (Part I, Section 1 and (6.1); Part II, (13.1)); x marks half the original length; E the ascent-first even series; e_m ; the gamma-ratio polynomials $G_r(a, b)$ (Part I, (7.7); Part II, (15.1)–(15.2), the same polynomials through the exponential recursion); Lambert W_{-1} . At $q = 1$, Part II’s $e_m(q)$, $r_m(q)$, $E(x, q)$, $R(x, q)$ are Part I’s e_m , $r_m = o_m$, $E(x)$, $O(x)$. The symbols below change meaning between the Parts; each Part’s text fixes its meaning there, and each Part opens with a short table of its own readings. No symbol was renamed.

Symbol	Meanings
$\mathcal{B}, B$	Part I: $\mathcal{B}_n$ the class; $B_j(z)$ the Bernoulli polynomials ((7.7)). Part II: $B_{s,\ell}(w)$ the half-power profiles; $\mathcal{B}_j(t)$ the Bernoulli polynomials. <i>Tempting false reading:</i> Part II’s calligraphic $\mathcal{B}_j$ is a Bernoulli polynomial, not Part I’s class $\mathcal{B}_n$. Section 23: Min’s B_n is $\mathcal{B}_n$.

Symbol	Meanings
R, r_m	Part I: $\mathcal{R}_n$ the reverse class, r_m the involutions in $\mathcal{R}_{2m}$. Part II: $R(x, q)$ the reverse series (Part I's O at $q = 1$), $r_m(q)$; R also a disc radius in Part I's Section 6.
O, o_m	Part I only: $O(x)$, $o_m = a_{2m+1}$. Part II writes the same object as R, r_m .
$A, \mathcal{A}$	Part I: $\mathcal{A}_s$ the amplitudes (constants), $A_s(x)$ analytic functions (6.7)–(6.8), $A = 1 \oplus \sigma \oplus 1$ a block. Part II: $A_{s,\ell,\varepsilon}(w)$ the integer-power profiles.
D	Part I: the discriminant $D(x) = M(x)P(x)$ (6.3); $D_{s,J}$ bracket constants. Part II: the Euler operator $w d/dw$ (14.3).
H	Part I: $H = O - 1$ (Section 6); $H_{s,K}$ bracket constants. Part II: $H(y) = (1 + y)/\sqrt{1 - y^2}$ (13.6).
P, Q	Part I: $P(x) = 1 - 2x - 4x^2$, P_0 in the quartic relation. Part II: the polynomials $P_\ell(r)$, $Q_\ell(w)$ (15.3)–(15.4).
ρ	Part I: the singularity $(\sqrt{5} - 1)/4$; ρ_- . Part II: a circle radius $\rho > 1$ in the proof of Theorem 20.1.
ε	Part I: $\varepsilon = 1/X_s$ (Section 8). Part II: the parity $\varepsilon = (-1)^m$.
y	Part I: the target value of the sequence. Part II: $y = 2x$.
t	Part I: $t = 1 - x/\rho$. Part II: $t = m^{-1/2}$ (Section 19), and other local variables.
K	Part I: $K_s(x)$ the singular amplitude functions, K an order. Part II: $K = \lceil \log m \rceil$ (Section 16); K the number of fixed-point pairs (Section 20).
b, d	Part I: $b = \log \Lambda$; $d_{s,j}$ the logarithmic coefficients. Part II: $b_N(\alpha)$ binomial coefficients, $b_s = B_{s,0}$; $d_r = C_r/(2 \cdot 4^r)$. Section 23: Min's b_n is a_n .
N, θ	Part I: $N_s(y)$, $N(y)$ the length thresholds. Part II: $N = m - 2r$; N an order in Theorem 19.1; $\theta_{s,m}$ the weight thresholds. <i>False reading</i> : Part II's “exact thresholds” are thresholds in the weight at fixed m , not Part I's thresholds in the length.
L, U, V	Part I: $L_s(y)$, $U_s(y)$ the bracket endpoints; $U_m = \mathcal{B}_{2m} $, $V_m = \mathcal{R}_{2m} $. Part II: $L_h(a, b)$ and the order L ; U, V factors in the proof of Lemma 17.1.
c, v, u, λ	Part I: $c_{s,j}$ the asymptotic coefficients, $\lambda_s = c_{s,1}$; $v_{s,k}$ the inverse coefficients, u the root shift. Part II: $c_\ell = 2^\ell G_\ell(1/2, 2)$ (15.8); u_j the interior inverse coefficients, $v_{n-1}(t)$ their partial sums; λ a real exponent in Theorem 20.1.
J, s, S	Part I: J an order; $s \in \{e, o\}$. Part II: J_r a binomial variable; $s \in \{E, R\}$; $S_\ell(r)$ polynomials and $S = \sqrt{2\pi}$.
W	Lambert W_{-1} (both). Part II also: W the radius of the window $ w \leq W$.

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243. Their author lines read “Report 149” and “Report 151”; neither names an author, a tool or an addressee (each sets an empty PDF author field), neither carries “prepared for private review” wording, and neither pins a ProveIt commit or names a repository path. They arrived in commit 60f54ea06 and were placed by 47fc7a069 (batch 106).

- **Part I**, bundle Report 149, the base, archive `Corrected_Baxter_Involutions_Asymptotics_and_Inverses_Source.zip` (496,609 bytes, 16 files; `Report149.tex`, 917 lines, 15 pages, dated 3 October 2026). Contributes the decomposition and the correction, the omitted involutions, the radicals, the fixed-order asymptotics, the length inverses and the ceiling brackets.
- **Part II**, bundle Report 151, archive `Baxter_Fixed_Point_Crossover_Profiles_and_Inverses_Source.zip` (912,824 bytes, 9 files, two of them Report 149's source and PDF under `foundation/`; `Report151.tex`, 977 lines, 17 pages, dated 3 October 2026). Contributes the

refinement by pairs of fixed points: the bivariate system and its factorizations, the uniform crossover theorem and its profile algorithms, the weight inverses and the limit laws.

Where the merge had to choose:

1. *Order and base.* Report 149 first, as the base: it is self-contained and carries the main finding; Report 151 imports its decomposition and has nothing on the length asymptotics.
2. *Printed once.* Report 151’s Section 1 opens with a paragraph restating Report 149’s definitions (alternation, involutions, the vincular form of the Baxter condition, the empty permutation) and a paragraph recalling Report 149’s decomposition, its sources (Min–Park, Ouchterlony) and the values 2168 and 2166 at length 20. Neither contains anything not in Part I; each is replaced by a **[write]** pointer, and the two sentences of the second paragraph that are Report 151’s own scope statements (“Setting $q = 1 \dots$ ” and the OEIS sentence) are printed as delivered. The definitions of C and M (13.1) and of G_ℓ (15.1)–(15.2) are kept, with notes, because Part II’s later text cites them by number. Part II’s Theorem 13.1 at $q = 1$ is Part I’s Theorem 1.1; its proof re-checks only how the fixed points sit in each block, and is kept.
3. *Cross-references.* Report 151’s text names its own Sections 5, 6 and 7 by number; they are printed as references to Sections 16, 17 and 18. Its citations of “Report 149” are Part I.
4. *Front matter and bibliography.* The two title blocks and abstracts, and Report 149’s “Scope” paragraph and Report 151’s, are printed at the heads of their Parts; one table of contents replaces two. The two bibliographies are merged, keeping every detail of every entry; where they annotate an entry differently (Min (2021), the OEIS entry, Min–Park), both annotations are kept. The write added the entries for Guibert–Linusson [11], Dokos–Pak [12] and the transseries volume [13].
5. *Write additions.* This front matter, the reading-conventions tables of the Parts, Section 23 with Remarks 23.1, 23.3, 23.4, 23.5 and Proposition 23.2, the dated notes, labels for unlabelled sections, and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 106, 5 October 2026) both manuscripts were read in full and no step was found false. Checked by hand there: Part I’s structural proof (Sections 3–4: the bound $r \leq 2k$, the form $A \ominus \eta \ominus A^{-1}$, the converse, and that there is no division by two); $D = M \cdot P$ from $M^2 = M - x^2$; that ρ is a root of $1 - 2x - 4x^2$; $[q]e_{10} = 126 = 2^9 h_{10}$; the expansion (18.2) of $\theta_{R,m}$ from h_m . Independent programs, written from scratch at intake, enumerated the alternating Baxter involutions by backtracking for $n \leq 22$ under the vincular definition and, separately, under classical avoidance of 2413 and 3142: both give 1, 1, 1, 1, 2, 2, 3, 5, 8, 12, 16, 32, 44, 84, 105, 231, 292, 636, 768, 1792, 2168, 5080, 6014 with identical fixed-point distributions, both witnesses of Section 5 are among the 2168, and both pass Min’s strict four-index definition. The bivariate recurrence of Theorem 13.1 equals these fixed-point distributions for every $n \leq 22$. Min’s recurrence, implemented literally, reproduces all 42 OEIS terms. The scaled residuals $m(\log s_m - m \log \Lambda + \frac{3}{2} \log m - \log \mathcal{A}_s)$ at $m = 1200$ are -0.71271 (even) and -2.62021 (odd), approaching λ_e and λ_o ; Part II’s boundary roots and leading profiles were checked numerically at $m \leq 321$ (note after Proposition 18.2). The delivered suites pass on copies, except checks of the POSIX-only output layers (README). The write re-read Min’s paper (publisher’s PDF, pages

255–257 rendered), the live OEIS entry and Barnabei et al.’s Section 10 (arXiv version), checked the two witnesses again by a direct program (involution, alternation, all 4845 quadruples of Min’s strict definition, outer blocks inverse to each other and not involutions), and checked that a_n is nondecreasing for $n \leq 2001$ and strictly increasing for $5 \leq n \leq 2001$ (from the corrected recurrence).

What was *not* done: the analytic estimates of Part II (factorial tails, the growing-shift remainder, the uniform remainders of the inverses and laws) were read and their displayed algebra spot-checked, but they are the manuscript’s proofs, corroborated numerically, not an independent verification. External inputs: Min–Park’s three necessity statements are accepted by Part I as published and were not re-proved by the write (Section 23, item 5); the write did not read Min–Park (2006), Ouchterlony (2006), Brignall–Huczynska–Vatter (2008), Park–Rizzolo (located at intake, arXiv:2512.25006v2, revised 1 September 2026) or Guibert–Linusson (2000), whose publisher’s page refused access; Guibert–Linusson’s statement is quoted from Min (2021, p. 254) and from Dokos and Pak’s abstract [12]. Those citations are as the manuscripts give them.

Relation to neighbouring reports and formal projects

No other report of the collection treats A347546, Baxter involutions or doubly alternating permutations (repository search, 5 October 2026; the word “Baxter” occurs elsewhere only for Rota–Baxter algebras and for A. M. Baxter’s work on ascent sequences). Related in subject, with no shared statement: `a113226-vincular-avoiders` (avoiders of the vincular pattern 12–34), `a239144-forbidden-distance-involutions` (statistics of involutions in another class), `enumerative-combinatorics/mesh-avoidance-catalan-inflation` (Catalan inflation for a mesh-avoidance class), and batch 105’s `a217057-unique-pattern-occurrences` and `a224182-unique-1432-order` (single pattern occurrences). The length inverses of Part I and the reverse leading inverse of Part II are instances of the transseries volume’s Lambert core (above); Part I’s ceiling brackets follow the pattern of its staircase theorem but are proved directly at the integers. Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats these permutations, and no statement of this report is formalized.

Part I

Corrected enumeration and asymptotics of alternating Baxter involutions

Bundle Report 149, archive `Corrected_Baxter_Involutions_Asymptotics_and_Inverses_Source.zip`, delivered as `Report149.tex` (15 pages), the base of the merge. Delivered title block “Corrected enumeration and asymptotics of alternating Baxter involutions / The class intended by OEIS A347546 and its parity inverses / Report 149 / 3 October 2026”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery.

Reading conventions of Part I	Elsewhere in the report
$\mathcal{B}_n, \mathcal{R}_n$ the classes; $a_n, e_m = a_{2m}, o_m = a_{2m+1};$ r_m the involutions in $\mathcal{R}_{2m}$ ($o_m = r_m$) $E(x), O(x); C, M, P(x) = 1 - 2x - 4x^2,$ $D = MP$ $\rho = (\sqrt{5} - 1)/4, \Lambda = 1 + \sqrt{5}, t = 1 - x/\rho$ $s \in \{e, o\}; \mathcal{A}_s, A_s(x), K_s(x), c_{s,j}, \lambda_s = c_{s,1}$ $b = \log \Lambda, p = 3/2, d_{s,j}; \varepsilon = 1/X_s; v_{s,k}$ y a target value; $N_s(y), N(y)$ thresholds in the length $B_j(z)$ Bernoulli polynomials; $G_r(a, b)$	Part II: $e_m(q), r_m(q)$, equal to these at $q = 1$; Min: $B_n, RB_n, b_n = a_n$ Part II: $R(x, q)$ for O ; D an Euler operator; $P_\ell(r)$ polynomials Part II: $\rho > 1$ a circle radius; $t = m^{-1/2}$ Part II: $s \in \{E, R\}; A_{s,\ell,\varepsilon}(w)$ profiles Part II: $\varepsilon = (-1)^m; b_N(\alpha), d_r; u_j$ Part II: $y = 2x; \theta_{s,m}$ thresholds in the weight Part II: $\mathcal{B}_j$ Bernoulli; the same G_ℓ

Abstract of this Part (as delivered). The alternating Baxter involutions named by OEIS A347546 are not counted by the recurrence printed in Min’s 2021 paper. The first incorrect term is at length 20: the class has 2168 objects, rather than 2166. The outside blocks of an involution are inverse partners; they need not individually be involutions. Starting from explicitly identified unrestricted structural lemmas of Min and Park (2006), we prove both directions and uniqueness of the corrected involution decomposition. Its outside factor is Catalan. The resulting even and odd half-length generating functions are algebraic, with exact dominant singularity $\rho = (\sqrt{5} - 1)/4$. We give exact leading amplitudes, explicit first corrections, an all-orders coefficient algorithm with fixed-order error bounds, and Lambert W_{-1} inverses for both parity subsequences. Ceiling-safe asymptotic brackets combine to at most two adjacent full indices, resolved by exact evaluation. Two explicit smallest omitted permutations and a newly written exact computational companion make the correction directly checkable.

(As delivered.) **Scope.** The substantive finding is the enumeration correction. Algebraic singularity transfer and formal inversion are standard tools applied to it. The report distinguishes the intended permutation class from the old recurrence-defined sequence throughout. The finite verification does not replace the all-length structural proof. No exhaustive literature-priority claim is made; Min’s 2002 thesis was not inspected. No external submission is part of this report.

1 Definitions and the corrected result

An alternating permutation begins with a rise: $\pi_1 < \pi_2 > \pi_3 < \dots$. Reverse alternation begins with a descent. A permutation is doubly alternating if both it and its inverse have the specified alternating convention. For an involution these two conditions coincide. The empty permutation is admitted.

We use the original strict-index Baxter definition: for every $1 \leq i < j < k < \ell \leq n$,

$$\pi_i + 1 = \pi_\ell < \pi_j \implies \pi_k > \pi_\ell, \quad (1.1)$$

$$\pi_\ell + 1 = \pi_i < \pi_k \implies \pi_j > \pi_i. \quad (1.2)$$

Equivalently, Baxter permutations avoid the vincular patterns 2-41-3 and 3-14-2: there are no $i < j < j+1 < \ell$ with, respectively,

$$\pi_{j+1} < \pi_i < \pi_\ell < \pi_j, \quad \pi_j < \pi_\ell < \pi_i < \pi_{j+1}.$$

Both forms are checked independently in the companion. Strict indices matter: a weak-index version printed in the recalled definition of Barnabei et al., Section 10 [5], is not used here.

Write $\mathcal{B}_n$ for the doubly ascent-first alternating Baxter class, and $\mathcal{R}_n$ for the doubly reverse-alternating Baxter class. Define

$$a_n = |\{\pi \in \mathcal{B}_n : \pi = \pi^{-1}\}|, \quad e_m = a_{2m}, \quad o_m = a_{2m+1}.$$

Thus $e_0 = o_0 = 1$. Put

$$E(x) = \sum_{m \geq 0} e_m x^m, \quad O(x) = \sum_{m \geq 0} o_m x^m.$$

$$C(x) = \sum_{j \geq 0} C_j x^j = \frac{1 - \sqrt{1 - 4x}}{2x}, \quad C_j = \frac{1}{j+1} \binom{2j}{j}.$$

The square root in this expression has constant term 1.

Theorem 1.1 (Corrected enumeration). *For every $m \geq 1$,*

$$o_m = \sum_{i=0}^{m-1} e_i o_{m-1-i}, \quad (1.3)$$

$$e_m = o_{m-1} + \sum_{j=0}^{\lfloor (m-2)/2 \rfloor} C_j e_{m-2j-2}. \quad (1.4)$$

An empty sum is zero. Equivalently,

$$O = 1 + xEO = \frac{1}{1 - xE}, \quad E = 1 + xO + x^2 EC(x^2). \quad (1.5)$$

These equations uniquely determine the two formal power series.

The first values of the corrected full sequence, indexed from $n = 0$, are

$$1, 1, 1, 1, 2, 2, 3, 5, 8, 12, 16, 32, 44, 84, 105, 231, 292, 636, 768, 1792, \\ 2168, 5080, 6014, 14594, 17252, 42204, 49360, 123120, 143672, 361288, \dots$$

The value at $n = 21$ is unchanged even though the value at $n = 20$ changes: the odd recurrence for o_{10} uses e_i only through $i = 9$.

Original length n	Corrected class a_n	Printed recurrence
19	1792	1792
20	2168	2166
21	5080	5080
22	6014	6012
23	14594	14592
24	17252	17234

The printed recurrence and the OEIS table inspected on 3 October 2026 [2, 3] agree with the class through $n = 19$. We use a_n only for the corrected class, and hats for the distinct old recurrence sequence when comparison is necessary.

[write] **The OEIS entry, 5 October 2026.** Read again by the write, A347546 is unchanged (revision #20 of June 29, 2022); the column “Printed recurrence” is its data. Proposition 23.2 proves that the recurrence values lie strictly below the class count at $n = 20$ and at every $n \geq 22$, and equal it otherwise; Remark 23.5 lists the 21 wrong terms, $n = 20$ and $n = 22, \dots, 41$. The corrected values agree for $n \leq 22$ with two direct enumerations written at intake (front matter, “What was checked”).

2 Published structural inputs and elementary closure

2.1 The precise source dependency

The proof uses the following statements from Min and Park’s original 2006 paper [1]. Its scanned printed pages 558–560 were checked directly. No involution restriction occurs in these statements.

1. Theorem 4.1: if $\pi \in \mathcal{B}_{2m}$, then $\pi_1 = 2k + 1$ for a unique $0 \leq k < m$, $\pi_{2m-2k} = 2m$, and the last $2k$ entries have values exactly $1, \dots, 2k$.
2. Corollary 4.5: if $\pi \in \mathcal{R}_{2m}$, then $\pi_1 = 2k$ for a unique $1 \leq k \leq m$, $\pi_{2k} = 1$, and the first $2k$ entries have values exactly $1, \dots, 2k$.
3. Corollary 4.7: every $\pi \in \mathcal{B}_{2m+1}$ begins with 1, and deletion of this entry followed by subtraction of 1 gives an element of $\mathcal{R}_{2m}$.

They are the unrestricted structural sources underlying Lemmas 2.1, 2.3, and 2.4 in Min [2]. We accept these published necessity lemmas as inputs and prove all required converses, inverse restrictions, and counting multiplicities below. We do not import Theorem 2.7 of that paper or its involution assertion about an outside block.

Ouchterlony [4] gives an independent structural route: his proceedings Corollaries 4.2–4.3 and Lemma 4.4 describe the block structure and identify the doubly alternating Baxter class with the doubly alternating class avoiding classical 2413 and 3142. The numbering is different in his thesis version. This is corroborating source evidence; the proof below states one complete set of inputs rather than switching silently between versions.

2.2 Direct sums and skew sums

For permutations α and β , $\alpha \oplus \beta$ puts all values of α below all values of β ; $\alpha \ominus \beta$ puts them above. Each block retains its internal order. The inverse identities are

$$(\alpha \oplus \beta)^{-1} = \alpha^{-1} \oplus \beta^{-1}, \quad (\alpha \ominus \beta)^{-1} = \beta^{-1} \ominus \alpha^{-1}. \quad (2.1)$$

The change of block order in the second identity is decisive.

Lemma 2.1. *Direct and skew sums of Baxter permutations are Baxter. A contiguous block whose values are a consecutive interval inherits the Baxter property upon standardization.*

Proof. Neither classical pattern 2413 nor 3142 has a proper initial segment consisting of its smallest or largest values. Thus both are indecomposable under either sum. A forbidden vincular occurrence crossing a sum boundary would make its underlying classical pattern decomposable, which is impossible. An occurrence within a component retains the required adjacent positions and is excluded there. For the second assertion, standardization of an interval block is a translation of values. Any violation inside it would be a violation in the original permutation. $\square$

This is not a claim that Baxter permutations are closed under arbitrary classical pattern extraction. Only the interval-block property just proved is needed.

3 The unrestricted Catalan factor

Before imposing involutionhood, the published inputs and closure give unique decompositions

$$\mathcal{B}_{2m} = \bigsqcup_{a+b=m-1} \{(1 \oplus \sigma \oplus 1) \ominus \tau : \sigma \in \mathcal{R}_{2a}, \tau \in \mathcal{B}_{2b}\}, \quad (3.1)$$

$$\mathcal{R}_{2m} = \bigsqcup_{a+b=m-1} \{(1 \ominus \tau \ominus 1) \oplus \sigma : \tau \in \mathcal{B}_{2a}, \sigma \in \mathcal{R}_{2b}\}. \quad (3.2)$$

We check both directions because the correct outside multiplicity must not be inferred from the disputed involution recurrence.

For (3.1), take $b = k$ from the first structural input. The first $2m - 2k$ entries form the upper interval, with its minimum first and maximum last. Deleting these two endpoints gives $\sigma \in \mathcal{R}_{2m-2k-2}$. Both position and value shifts are one, so the forward and inverse alternating conventions switch. The suffix has even position and value shifts and remains in $\mathcal{B}_{2k}$. The interval-block argument proves Baxter avoidance. The first value recovers k , hence every block and its standardization uniquely.

Conversely $A = 1 \oplus \sigma \oplus 1$ and A^{-1} are ascent-first alternating: their endpoint comparisons are rises and their interiors have the shifted reverse convention. Both A and τ have even length. The skew boundary is a descent at an even position, as required. Inversion reverses these two even blocks, so the same boundary check applies to $\tau^{-1} \ominus A^{-1}$. Lemma 2.1 gives the Baxter property.

For (3.2), the second structural input identifies the initial interval $[2k]$, with maximum first and minimum last. Deleting its two endpoints gives $\tau \in \mathcal{B}_{2k-2}$, and the suffix belongs to $\mathcal{R}_{2m-2k}$. The one-position and one-value shifts change both interior conventions; the even suffix shifts preserve them. The first value again recovers k . Conversely the skew endpoint boundaries are descents at odd positions, and the direct boundary after the even initial component is a rise at an even position.

Inversion preserves the direct-block order, so the same checks prove double reverse alternation and closure proves Baxter avoidance.

Let $U_m = |\mathcal{B}_{2m}|$ and $V_m = |\mathcal{R}_{2m}|$, with $U_0 = V_0 = 1$. Their generating functions consequently satisfy

$$U = 1 + xVU, \quad V = 1 + xUV.$$

Subtracting gives $U = V$, hence $U = V = C$, the unique formal solution of $C = 1 + xC^2$ with constant term 1. In particular,

$$|\mathcal{R}_{2j}| = |\mathcal{B}_{2j}| = C_j. \quad (3.3)$$

Every member in these two decompositions is recursively built from direct and skew sums of singletons. Thus it avoids classical 2413 and 3142 as well. This also makes inverse closure transparent for these particular classes. The result is consistent with the class identification cited by Barnabei et al. [5], without using their cited involution recurrence.

[write] Prior art: the Catalan count, 5 October 2026. The count (3.3) is not new. Guibert and Linusson [11] proved that the doubly alternating Baxter permutations of length $2n$ and of length $2n + 1$ are counted by C_n , and Min and Park [1] proved it again; Min [2, p. 254] cites both for $|\mathcal{B}_{2n+\delta}| = C_n$ ($\delta = 0, 1$). The reverse class follows from the odd case: $|\mathcal{R}_{2j}| = |\mathcal{B}_{2j+1}| = C_j$ by the third structural input with direct-sum closure (the bijection $\sigma \mapsto 1 \oplus \sigma$ of Subsection 4.2). Report 149 does not cite Guibert–Linusson. This section re-derives their count from Min–Park’s necessity lemmas: it is a second route to a published theorem, which Part I needs in the form (3.1)–(3.2), not a new count. The new enumeration is the involution count of Section 4. (Guibert–Linusson’s paper itself was not read by the write; front matter.)

4 The involution restriction and its completeness

4.1 Even lengths

Let $\pi \in \mathcal{B}_{2m}$ be an involution and write $\pi_1 = 2k + 1$. Put $r = 2m - 2k$. If $k = 0$, then $\pi = 1 \oplus \sigma \oplus 1$ and inversion fixes both endpoints. Thus σ must be an involution in $\mathcal{R}_{2m-2}$.

Suppose $k > 0$. The value 1 lies in the last $2k$ positions. Involutionhood gives $\pi(2k + 1) = 1$, so $2k + 1$ lies among these positions. Hence $r \leq 2k$, or $k \geq \lceil m/2 \rceil$. The first r positions map onto the last r values. Their inverse images force the last r positions to map back to the first r values, and the center interval is invariant. Standardization gives

$$\pi = A \ominus \eta \ominus A^{-1}, \quad A = 1 \oplus \sigma \oplus 1, \quad (4.1)$$

where

$$\sigma \in \mathcal{R}_{2j}, \quad j = m - k - 1, \quad \eta \in \mathcal{B}_{2(m-2j-2)}, \quad \eta = \eta^{-1}, \quad 0 \leq j \leq \lfloor (m-2)/2 \rfloor.$$

The outside blocks are inverse partners because the full permutation is an involution. They are not required to coincide. Their forward alternation and inverse pairing imply that A is doubly alternating; deleting its endpoints gives the unrestricted σ above. The center has even shifts and is an alternating Baxter involution.

Conversely, choose any such σ and η , including the empty center when $m = 2j + 2$. The construction (4.1) is Baxter by closure. The even block lengths put all skew boundaries at even positions, so they are the required descents. The interior and endpoint checks from (3.1) give the remaining alternation. Finally,

$$(A \ominus \eta \ominus A^{-1})^{-1} = A \ominus \eta^{-1} \ominus A^{-1} = A \ominus \eta \ominus A^{-1}.$$

Its first value is $2(m - j - 1) + 1$, which recovers j . Standardizing the recovered intervals then recovers σ and η . This proves sufficiency and uniqueness.

There are C_j outside choices by (3.3). There is no division by two: the left block has a distinguished position. Choosing σ^{-1} on the left usually gives a different full permutation. With r_m denoting the involutions in $\mathcal{R}_{2m}$, we have proved

$$e_m = r_{m-1} + \sum_{j=0}^{\lfloor (m-2)/2 \rfloor} C_j e_{m-2j-2}.$$

4.2 Odd lengths

The third structural input, together with direct-sum closure, gives a bijection $\sigma \mapsto 1 \oplus \sigma$ from $\mathcal{R}_{2m}$ onto $\mathcal{B}_{2m+1}$. Its inverse preserves involutionhood, so $o_m = r_m$. In (3.2), both the first component and its suffix map their own position intervals to themselves. Inversion preserves their order. They cannot exchange as the outside skew blocks did. Thus $(1 \ominus \tau \ominus 1) \oplus \sigma$ is an involution exactly when both τ and σ are involutions. This proves

$$o_m = \sum_{i=0}^{m-1} e_i o_{m-1-i}.$$

Together with the even result, this proves Theorem 1.1. Every right-hand coefficient uses strictly smaller indices, proving formal uniqueness as well.

4.3 A canonical block interpretation

The block $1 \oplus \sigma \oplus 1$ is skew-indecomposable because its first value is its minimum. Iterating (3.1) gives the canonical skew sequence of these primitives. Inversion reverses the sequence and inverts each primitive. An involution therefore consists of any number of inverse-paired outside primitives and an optional involutive central primitive. A pair has weight $x^2 C(x^2)$; the central choices have weight $1 + xO$. This gives

$$E = \frac{1 + xO}{1 - x^2 C(x^2)}.$$

Likewise $1 \ominus \tau \ominus 1$ is direct-indecomposable, and inversion preserves the direct sequence. Each primitive must separately be an involution, giving $O = 1/(1 - xE)$. This restates the same complete proof while isolating the source of the different restrictions.

5 The first missing objects and the published error

Min's proof [2, p. 256] asserts that the standardized first outside block belongs to the involution subclass and identifies it with the last outside block. The correct assertion is

$$\text{std}(\pi_3) = \text{std}(\pi_1)^{-1}.$$

Only the middle block is invariant by itself. Replacing the unrestricted C_j in (1.4) by the involution count o_j is the counting error. In hat notation the printed equations instead define

$$\hat{E} = 1 + x\hat{O} + x^2\hat{E}\hat{O}(x^2), \quad \hat{O} = (1 - x\hat{E})^{-1}.$$

They are meaningful formal equations for another sequence, but do not count the stated class beyond the agreement range.

Now $C_j = o_j$ for $0 \leq j \leq 3$, while $C_4 = 14$ and $o_4 = 12$. The first place where the difference can occur in the even recurrence is $m = 10$, $j = 4$, with empty center. Lower-index recurrence terms still agree. Thus the discrepancy of exactly two at length 20 is structural, not just a computational lower bound.

The two non-involutive interiors in $\mathcal{R}_8$ are the inverse pair

$$\sigma = (8, 4, 6, 5, 7, 2, 3, 1), \quad \sigma^{-1} = (8, 6, 7, 2, 4, 3, 5, 1).$$

Set $A = 1 \oplus \sigma \oplus 1$. Placing either orientation first in $A \ominus A^{-1}$ gives the omitted permutations

$$\begin{aligned} \pi^{(1)} &= (11, 19, 15, 17, 16, 18, 13, 14, 12, 20, \\ &\quad 1, 9, 7, 8, 3, 5, 4, 6, 2, 10), \\ \pi^{(2)} &= (11, 19, 17, 18, 13, 15, 14, 16, 12, 20, \\ &\quad 1, 9, 5, 7, 6, 8, 3, 4, 2, 10). \end{aligned}$$

Both satisfy the original strict-index Baxter definition, the standard vincular definition, alternation, and involutionhood. The companion checks every one of the $\binom{20}{4} = 4845$ strict quadruples for each witness, as well as all positional-vincular possibilities. Their standardized outside blocks are inverse and unequal. Finite direct checks establish their membership; the decomposition proves that they account for the entire first discrepancy.

[write] **The published proof, 5 October 2026.** Min writes $\pi = \pi_1 \pi_2 \pi_3$, with π_1^* the standardized first block, and asserts “ $\pi_1^* \in B_{2n-2k}(I)$ ” and “Since π is an involution, $\pi_1^* = \pi_3$ ” [2, p. 256]. Section 23 quotes her Theorem 2.7 and this step (Remark 23.1), proves that the odd formula is right and the even formula wrong exactly from $n = 10$ on, with the recurrence values below the class at $n = 20$ and every $n \geq 22$ (Proposition 23.2), checks the two permutations above against her own definition (Remark 23.3), and records her printed list and the data of A347546 (Remarks 23.4 and 23.5).

6 Algebraic generating functions and the dominant singularity

Set

$$M(x) = 1 - x^2 C(x^2) = \frac{1 + \sqrt{1 - 4x^2}}{2}, \quad P(x) = 1 - 2x - 4x^2. \quad (6.1)$$

Eliminating E from (1.5) gives

$$x^2 O^2 - (M - x)O + M = 0. \quad (6.2)$$

Since $M^2 - M + x^2 = 0$, its discriminant factors as

$$D(x) = (M - x)^2 - 4Mx^2 = M(x)P(x). \quad (6.3)$$

The branches analytic at zero with constant terms 1 are

$$O(x) = \frac{M(x) - x - \sqrt{D(x)}}{2x^2}, \quad E(x) = \frac{M(x) + x - \sqrt{D(x)}}{2xM(x)}. \quad (6.4)$$

Their apparent singularities at zero are removable. The radical series and recurrence coefficients are computed by independent exact algorithms in the companion.

For an explicit polynomial relation define $P_0 = x^2 O^2 - (1 - x)O + 1$ and $H = O - 1$. Then

$$P_0^2 + H P_0 + x^2 H^2 = 0. \quad (6.5)$$

Indeed $P_0 = -x^2 C(x^2)(O - 1)$, and the Catalan equation gives this identity. Equivalently,

$$[2x^2 O^2 - (1 - 2x)O + 1]^2 - (1 - 4x^2)(O - 1)^2 = 0.$$

The specified analytic branch is essential; a polynomial equation alone does not select the counting series.

Let

$$\rho = \frac{\sqrt{5} - 1}{4}, \quad \rho_- = -\frac{1 + \sqrt{5}}{4}, \quad \Lambda = \rho^{-1} = 1 + \sqrt{5}. \quad (6.6)$$

Then $P(x) = (1 - x/\rho)(1 - x/\rho_-)$. The function M is analytic and nonzero for $|x| < 1/2$: its square root is the branch equal to 1 at zero, and $M = 0$ would force $x = 0$, where $M = 1$. The only zero of P in that disk is ρ , since $|\rho_-| > 1/2$. It is simple, with $P'(\rho) = -2\sqrt{5}$. Thus the two series have a unique dominant singularity at ρ , with a nonzero square-root amplitude. For any R strictly between ρ and $1/2$, their formulas continue to a disk of radius R slit along the positive real ray from ρ . This gives the required Delta domain for coefficient transfer.

More explicitly, put $t = 1 - x/\rho$ and

$$K_o(x) = \frac{\sqrt{M(x)(1 - x/\rho_-)}}{2x^2}, \quad A_o(x) = \frac{M(x) - x}{2x^2}, \quad (6.7)$$

$$K_e(x) = \frac{x K_o(x)}{M(x)}, \quad A_e(x) = \frac{M(x) + x}{2x M(x)}. \quad (6.8)$$

In a neighborhood of ρ , all four amplitudes are analytic and

$$O(x) = A_o(x) - K_o(x)t^{1/2}, \quad E(x) = A_e(x) - K_e(x)t^{1/2}. \quad (6.9)$$

The amplitudes here need not be analytic at zero individually; the original combinations are. Only their local analyticity at ρ is used for singularity analysis.

7 Leading terms and every fixed asymptotic order

7.1 Exact leading constants and first corrections

Let $M_\rho = M(\rho) = (1 + \sqrt{2\rho})/2$. Define

$$\mathcal{A}_o = \frac{\sqrt{2\sqrt{5}\rho M_\rho}}{4\sqrt{\pi}\rho^2}, \quad \mathcal{A}_e = \frac{\sqrt{2\sqrt{5}\rho M_\rho}}{4\sqrt{\pi}\rho M_\rho}. \quad (7.1)$$

The symbols $\mathcal{A}_s$ are constants, distinguished from the analytic functions $A_s(x)$ in (6.7)–(6.8).

Theorem 7.1. *For $s \in \{e, o\}$ and every fixed integer $J \geq 0$, there are explicit real constants $c_{s,j}$ such that*

$$s_m = \mathcal{A}_s \Lambda^m m^{-3/2} \left(1 + \sum_{j=1}^J c_{s,j} m^{-j} + O_{s,J}(m^{-J-1}) \right). \quad (7.2)$$

Here $s_m = e_m$ or o_m according to s . The first constants $\lambda_s = c_{s,1}$ are

$$\lambda_o = -\frac{21}{8} + \frac{3\rho}{8} \frac{D''(\rho)}{D'(\rho)}, \quad (7.3)$$

$$\lambda_e = \lambda_o + \frac{3}{2} - \frac{3\rho}{2} \frac{M'(\rho)}{M(\rho)}. \quad (7.4)$$

The error assertion is for each fixed order, not uniform in J .

Numerically, these constants are

$$\begin{aligned} \rho &= 0.30901699437494742410\dots, & \Lambda &= 3.23606797749978969641\dots, \\ \mathcal{A}_o &= 1.64094172716863543428\dots, & \mathcal{A}_e &= 0.56778936745075026465\dots, \\ \lambda_o &= -2.62171983544803595042\dots, & \lambda_e &= -0.71369036117693250404\dots \end{aligned}$$

These decimals are informative approximations; the displayed exact expressions are the definitions. Per original permutation entry the exponential rate is $\sqrt{\Lambda} = 1.79890743994786727226\dots$

Proof. The constant multiplying $-t^{1/2}$ is $K_s(\rho)$, and $-1/\Gamma(-1/2) = 1/(2\sqrt{\pi})$, giving $\mathcal{A}_s = K_s(\rho)/(2\sqrt{\pi})$. Since $1 - \rho/\rho_- = 2\sqrt{5}\rho$, these are precisely (7.1). Expand $K_s(\rho(1-t)) = k_{s,0} + k_{s,1}t + \dots$, with $k_{s,1} = -\rho K'_s(\rho)$. The first gamma-ratio correction to $t^{1/2}$ is $3/(8m)$ and $\Gamma(-1/2)/\Gamma(-3/2) = -3/2$. Therefore

$$\lambda_s = \frac{3}{8} - \frac{3}{2} \frac{k_{s,1}}{k_{s,0}} = \frac{3}{8} + \frac{3\rho}{2} \frac{K'_s(\rho)}{K_s(\rho)}. \quad (7.5)$$

Using $K_o = \sqrt{D/(1-x/\rho)/(2x^2)}$ gives $K'_o/K_o = D''(\rho)/(4D'(\rho)) - 2/\rho$ at ρ . This proves (7.3); logarithmic differentiation of $K_e = xK_o/M$ proves (7.4). The next subsection establishes the full expansion and remainder. $\square$

As an alternative exact simplification, using $M'(\rho) = -2\rho/\sqrt{2\rho}$,

$$\lambda_o = -\frac{21}{8} + \frac{3\rho}{4} \frac{M'(\rho)}{M_\rho} + \frac{3\rho}{2\sqrt{5}}, \quad \lambda_e = -\frac{9}{8} - \frac{3\rho}{4} \frac{M'(\rho)}{M_\rho} + \frac{3\rho}{2\sqrt{5}}.$$

7.2 A coefficient algorithm with a rigorous remainder

Define for $j \geq 0$

$$k_{s,j} = [t^j]K_s(\rho(1-t)).$$

For every fixed $K \geq 1$, the Delta-domain transfer theorem applied to (6.9) gives

$$s_m = -\rho^{-m} \sum_{j=0}^{K-1} k_{s,j} \frac{\Gamma(m-j-1/2)}{\Gamma(-j-1/2)\Gamma(m+1)} + O_{s,K}(\rho^{-m}m^{-K-3/2}). \quad (7.6)$$

The local analytic part contributes no algebraic-order terms to this expansion. The next nonanalytic power is $t^{K+1/2}$; the analytic continuation established above gives the stated coefficient remainder, by the standard transfer theorem [7, Chapter VI]. Equivalently one may retain one further local term before applying a remainder estimate. No singularity of equal modulus has been omitted.

For a fully specified inverse-power algorithm, let $B_j(z)$ denote the Bernoulli polynomials, normalized by $te^{zt}/(e^t - 1) = \sum_{j \geq 0} B_j(z)t^j/j!$. For fixed a, b define $G_r(a, b)$ by

$$\sum_{r \geq 0} G_r(a, b)z^r = \exp \left(\sum_{\ell \geq 1} \frac{(-1)^{\ell+1}(B_{\ell+1}(a) - B_{\ell+1}(b))}{\ell(\ell+1)} z^\ell \right). \quad (7.7)$$

Then the fixed-order gamma-ratio expansion is

$$\frac{\Gamma(m+a)}{\Gamma(m+b)} \sim m^{a-b} \sum_{r \geq 0} G_r(a, b)m^{-r}.$$

Put $c_{s,0} = 1$. Direct substitution into (7.6) gives the finite formula

$$c_{s,q} = -\frac{1}{\mathcal{A}_s} \sum_{j=0}^q \frac{k_{s,j}}{\Gamma(-j-1/2)} G_{q-j}(-j-1/2, 1), \quad q \geq 0. \quad (7.8)$$

Thus every coefficient is determined by derivatives of explicit radicals at the quadratic irrational ρ and rational Bernoulli-polynomial operations. Formula (7.6) with $K = J + 1$ proves (7.2). All relative coefficients $c_{s,q}$ are algebraic; the common $\sqrt{\pi}$ factors cancel. We claim a full fixed-order Poincare expansion, not convergence of its infinite formal series.

7.3 Parity and the full generating function

The full ordinary generating function is

$$\mathcal{G}(z) = E(z^2) + zO(z^2).$$

It is algebraic, hence D-finite. General algebraicity for alternating separable involutions also follows from the earlier theorem of Brignall, Huczynska, and Vatter [6, Corollary 1.4 and Theorem 5.3]: the alternating and involution restrictions can be combined in a class with finitely many simple permutations. Applying that theorem to the separable class is an independent inference, so algebraicity itself is not a new claim of this report. The corrected explicit equations identify the relevant branch and coefficients. Its dominant singularities are $z = \pm\sqrt{\rho}$; neither leading amplitude cancels, since $K_e(\rho)/K_o(\rho) = \rho/M_\rho \neq \sqrt{\rho}$. The parity-specific formulas avoid cancellation between their two contributions. In particular,

$$a_{2m} \sim \mathcal{A}_e \Lambda^m m^{-3/2}, \quad a_{2m+1} \sim \mathcal{A}_o \Lambda^m m^{-3/2}.$$

There is no need for logarithmic or log-periodic leading factors. The two amplitudes cannot be merged into one amplitude independent of parity. Their consequences include

$$\frac{o_m}{e_m} \longrightarrow \frac{M_\rho}{\rho} > 1, \quad \frac{e_{m+1}}{o_m} \longrightarrow \frac{1}{M_\rho} > 1. \quad (7.9)$$

Thus the full sequence is eventually strictly increasing. This is an asymptotic conclusion, not a claimed all-index monotonicity proof. No claim concerning a natural boundary or non-D-finiteness of the old recurrence sequence is needed for the corrected result.

[write] **Monotonicity and the old sequence, 5 October 2026.** From the corrected recurrence, a_n is nondecreasing for $n \leq 2001$ and strictly increasing for $5 \leq n \leq 2001$ ($a_0 = a_1 = a_2 = a_3$ and $a_4 = a_5$); an all-index proof is open (Section 23, item 2), and so is the nature of the old sequence (item 3).

8 Lambert core and every fixed inverse order

8.1 Large real model inverses

Let $p = 3/2$ and $b = \log \Lambda > 0$. Define logarithmic coefficients $d_{s,j}$ formally by

$$\log \left(1 + \sum_{j \geq 1} c_{s,j} z^j \right) = \sum_{j \geq 1} d_{s,j} z^j. \quad (8.1)$$

For example $d_{s,1} = c_{s,1}$, $d_{s,2} = c_{s,2} - c_{s,1}^2/2$, and $d_{s,3} = c_{s,3} - c_{s,1}c_{s,2} + c_{s,1}^3/3$. Equation (7.2) implies, for fixed J ,

$$\log s_m = bm - p \log m + \log \mathcal{A}_s + \sum_{j=1}^J d_{s,j} m^{-j} + O_{s,J}(m^{-J-1}). \quad (8.2)$$

For large y , the exact large root of the leading real model is

$$X_s(y) = -\frac{p}{b} W_{-1} \left(-\frac{b}{p} \left(\frac{\mathcal{A}_s}{y} \right)^{1/p} \right). \quad (8.3)$$

The real branch W_{-1} is required. It selects $X_s > p/b$ and satisfies $\mathcal{A}_s e^{bX_s} X_s^{-p} = y$. The other real branch gives the small root, unrelated to inversion of the growing sequence. For sufficiently large y the displayed argument lies in $(-1/e, 0)$.

For fixed J put

$$F_{s,J}(x) = bx - p \log x + \log \mathcal{A}_s + \sum_{j=1}^J d_{s,j} x^{-j}.$$

Its derivative is $b - p/x + O(x^{-2}) \geq b/2$ for all sufficiently large x . Therefore $F_{s,J}(x) = \log y$ has a unique large real root $x_{s,J}(y)$. These are explicitly defined model inverses; no unspecified real interpolation of the integer sequence is assumed.

8.2 Formal recursion and residual proof

Set $x = X_s + u$, $\varepsilon = 1/X_s$. Subtraction of the exact leading root equation gives the formal equation

$$0 = bu - p \log(1 + \varepsilon u) + \sum_{j \geq 1} d_{s,j} \varepsilon^j (1 + \varepsilon u)^{-j}. \quad (8.4)$$

Its coefficient of u at $(\varepsilon, u) = (0, 0)$ is $b \neq 0$, so there is a unique formal solution $u = \sum_{k \geq 1} v_{s,k} \varepsilon^k$. Coefficient extraction gives, suppressing the subscript s ,

$$\begin{aligned} v_1 &= -d_1/b, \\ v_2 &= (pv_1 - d_2)/b, \\ v_3 &= (pv_2 + d_1v_1 - d_3)/b, \\ v_4 &= (pv_3 - (p/2)v_1^2 + d_1v_2 + 2d_2v_1 - d_4)/b. \end{aligned} \quad (8.5)$$

Each further equation is linear in the next coefficient, with leading coefficient b .

Theorem 8.1 (Fixed-order inverse expansion). *For each fixed $K \geq 1$ and $J \geq K$,*

$$x_{s,J}(y) = X_s(y) + \sum_{k=1}^K v_{s,k} X_s(y)^{-k} + O_{s,J,K}(X_s(y)^{-K-1}). \quad (8.6)$$

For $K = 0$, $x_{s,J} = X_s + O(X_s^{-1})$.

Proof. Substitution of the truncated formal solution into $F_{s,J}(X_s + u) - \log y$ cancels coefficients through order X_s^{-K} and leaves $O(X_s^{-K-1})$. All expansions are on a fixed small neighborhood of $u = 0$ for sufficiently large X_s . There $F'_{s,J} \geq b/2$. The mean value theorem converts the residual to a root error of the same order; existence and uniqueness follow from the eventual monotonicity of the model. For $K = 0$ the uncorrected residual is $O(X_s^{-1})$. No convergence assumption on the infinite formal series is used. $\square$

The limiting model derivative is a positive constant. Consequently inversion here supplies no extra power of X_s^{-1} beyond the residual order. Since $d_{s,1} = \lambda_s < 0$, the first inverse correction $v_{s,1}/X_s = -\lambda_s/(bX_s)$ is positive.

[write] **The transseries volume, 5 October 2026.** The centre (8.3) is the large solution of $aX + b' \log X = L$ with $a = \log \Lambda$ (this section's b), $b' = -p = -3/2$ and $L = \log(y/\mathcal{A}_s)$: it is an instance of `p0:thm:lambert-core(3)` of the transseries volume [13], whose branch rule for $b' < 0$ gives $X = (p/b)(-W_{-1}(-(b/p)e^{-L/p}))$, with $e^{-L/p} = (\mathcal{A}_s/y)^{1/p}$, and requires $L > p(1 - \log(p/b))$, the condition that the argument lies in $(-1/e, 0)$. With $x = X_s + u$ and $\varepsilon = 1/X_s$, (8.4) is the volume's `p0:eq:formal-exactness` with $q_j = d_{s,j}$, so $v_{s,k}$ are the coefficients c_k of `p0:thm:lambert-centered`; and Theorem 8.1 is part (5) of that theorem applied to $\exp F_{s,J}$, which is differentially of exponential-power type (`p0:def:model`) with data $(\mathcal{A}_s, \log \Lambda, 1, -3/2, 1, \sum_{j \leq J} d_{s,j} t^j)$, for orders $K \leq J$ (the coefficients c_k with $k \leq J$ involve only $d_{s,1}, \dots, d_{s,k}$).

9 Ceiling-safe thresholds and exact parity choice

Define the exact thresholds for $y > 0$ by

$$N_e(y) = \min\{m \geq 0 : e_m \geq y\}, \quad N_o(y) = \min\{m \geq 0 : o_m \geq y\}. \quad (9.1)$$

They exist since both subsequences grow without bound. Their eventual strict increase follows either from (7.2) or (7.9). In each statement below y is also larger than all finitely many values before the chosen monotonicity onset; no early crossing is silently ignored.

Theorem 9.1 (Safe asymptotic ceilings). *For every fixed $J \geq 0$ there exist $D_{s,J} > 0$ and an onset such that, for all sufficiently large y ,*

$$\left\lceil x_{s,J} - D_{s,J} x_{s,J}^{-J-1} \right\rceil \leq N_s(y) \leq \left\lceil x_{s,J} + D_{s,J} x_{s,J}^{-J-1} \right\rceil. \quad (9.2)$$

For $K \geq 1$ set $\hat{x}_{s,K} = X_s + \sum_{k=1}^K v_{s,k} X_s^{-k}$. There are constants $H_{s,K} > 0$ such that

$$L_s(y) = \left\lceil \hat{x}_{s,K} - H_{s,K} X_s^{-K-1} \right\rceil \leq N_s(y), \quad (9.3)$$

$$N_s(y) \leq U_s(y) = \left\lceil \hat{x}_{s,K} + H_{s,K} X_s^{-K-1} \right\rceil. \quad (9.4)$$

For $K = 0$ use $\hat{x}_{s,0} = X_s$ and half-width $H_{s,0}/X_s$.

Proof. Write $x = x_{s,J}$. At integers within two of x , the log-error in (8.2) is bounded by a constant times x^{-J-1} , while $F'_{s,J} \geq b/2$. Choose D with enough slack that integers $m < x - Dx^{-J-1}$ in this two-unit window satisfy $\log s_m < \log y$, and the integer $m = \lceil x + Dx^{-J-1} \rceil$ satisfies $\log s_m \geq \log y$. For integers outside that window, eventual monotonicity extends the lower comparison, and the finite pre-onset range is excluded by increasing y . This proves (9.2). Taking $J = K$ and absorbing the root error of (8.6) yields (9.3)–(9.4). For $K = 0$, the uncorrected root and log-error estimates both have order X_s^{-1} . $\square$

The use of $\geq$ in (9.1) is important. The result includes inputs equal to exact sequence values. Taking the ceiling of an approximate real inverse without its error interval would not be justified, however small the approximation error.

The full first-crossing index is exactly

$$N(y) := \min\{n \geq 0 : a_n \geq y\} = \min\{2N_e(y), 2N_o(y) + 1\}. \quad (9.5)$$

This identity requires no monotonicity between parities. The bounds above therefore give

$$\min\{2L_e, 2L_o + 1\} \leq N(y) \leq \min\{2U_e, 2U_o + 1\}. \quad (9.6)$$

Corollary 9.2. *For every fixed inverse order, the combined interval in (9.6) eventually consists of at most two adjacent full indices. If the bounds coincide they determine $N(y)$ exactly. Otherwise exact evaluation at the candidates resolves the threshold.*

Proof. Since $\mathcal{A}_o/\mathcal{A}_e = M_\rho/\rho$, subtraction of the two model equations, whose derivative tends to b , gives

$$x_{e,J}(y) - x_{o,J}(y) \longrightarrow \delta = \frac{\log(M_\rho/\rho)}{\log \Lambda} = 1 + \frac{\log M_\rho}{\log \Lambda}. \quad (9.7)$$

The same limit holds for the finite approximants. Because $\rho < M_\rho < 1$, we have $0 < \delta < 1$; numerically $\delta = 0.9037058146841100247\dots$. The error half-widths tend to zero. Both parity intervals cannot simultaneously contain an integer where their endpoint ceilings differ: that would force their center difference to approach an integer, contradicting (9.7). If neither interval has distinct endpoint ceilings, both parity thresholds are fixed and so is (9.5). If only the even interval has possible ceilings $m, m+1$, its center is $m + o(1)$; the odd center is $m - \delta + o(1)$ and its ceiling is m . The full candidates are $2m$ and $2m+1$. If only the odd interval has possible ceilings $m, m+1$, the even center is $m + \delta + o(1)$ with ceiling $m+1$, and the full candidates are $2m+1$ and $2m+2$. Endpoint equality is included by the use of closed bounds and ceilings. $\square$

The constants $D_{s,J}, H_{s,K}$ and their onsets are existence constants, not certified finite-input numerical values. The formulas are rigorous asymptotic localization theorems. The exact recurrence provides a terminating threshold algorithm at any finite input; the asymptotic brackets are a proven accelerator only once explicit bounds and onsets are additionally supplied. Numerical residual tests cannot turn unknown error constants into universal certificates.

[write] **Staircase, 5 October 2026.** The brackets (9.2)–(9.4) have the two-ceiling form of the separation condition `p0:thm:staircase(2)` of the transseries volume, and (9.5) has the form of its residue-class part (4) with $r = 2$. They are *not* instances of that theorem, which presupposes an admissible interpolation of the sequence: the proofs above work directly at the integers, and (9.5) needs no monotonicity at all. Corollary 9.2 has no counterpart in the volume. Effective constants and onsets are open (Section 23, item 1).

10 Computational evidence and reproducible scope

The source package contains a newly written exact companion, with no imported code from earlier reports. Its independent routes compare the corrected coefficient recurrence with formal expansion of the radicals (6.4), verify the functional and polynomial identities, locate the first disagreement with the old recurrence, and check the two explicit omitted permutations by both Baxter predicates. It also performs a bounded exhaustive permutation check and exact formal inverse-residual tests. The companion README and deterministic JSON specify each finite bound, coefficient range, and schema.

All exact tests are active in both ordinary Python and optimized `python3 -O`; no mathematical validation depends on removable `assert` statements. The semantic fixture checks reject altered values and malformed data. Output creation refuses overwriting and symlink destinations. These properties concern reproducibility and failure detection, not a computational proof of the infinite theorems.

Separate direct involution enumerations performed during the research checked the strict value-adjacency definition and the positional-vincular definition independently through original length 22. They give 2168, 5080, and 6014 at lengths 20, 21, and 22. Those corroborating results are distinguished in the provenance file from the smaller checks freshly rerun by the packaged companion. The report’s all-length result rests on the proved decomposition, not extrapolation of either finite list.

The PDF and ZIP builders use fixed inputs, normalized metadata, no shell escape, and new output paths. The ZIP allowlist excludes working notes, downloaded papers, unrelated report foundations, and intermediate build files. The source provenance file records the reviewed primary documents, their hashes, and the precise theorem locations used. Public source URLs remain the proper route to those papers. The final package supplies the report source, report PDF, companion, and rebuild instructions.

[write] **The companion here, 5 October 2026.** Report 149’s companion and release tools are shipped as `149-enum-companion-README.md`, `code/149-enum-*.py`, `data/149-enum-companion-evidence.json` and `data/149-enum-SOURCE_PROVENANCE.json`; the PDF, the archive’s checksum list and the ZIP it builds are not shipped. The report README gives the rerun route from the arrival commit, and which checks need a POSIX host.

11 Source review and limits of the claim

The original 2006 structural statements and the full 2021 paper were inspected, including the scanned original pages containing the inputs and the page containing the erroneous involution restriction. The Ouchterlony and Barnabei comparisons support the intended class identification. Barnabei’s citation of Min is not an independent proof of Min’s recurrence. Bounded searches of corrected parity prefixes, alternating separable involutions, and related symmetric-tree formulations found no matching explicit corrected enumeration in the sources checked. General algebraicity was already available from [6]. These negative search results do not establish global novelty. Min’s 2021 paper says that its material derives from the author’s 2002 thesis; that thesis remains uninspected.

The conclusions have distinct levels. The corrected enumeration is proved from explicit published unrestricted structural inputs. The radical formulas, exact singularity, fixed-order asymptotics, and inverse-localization statements follow analytically. Finite computations independently test their

algebra and the first omitted objects. The numerical decimals are approximations; the exact formulas specify the constants. Effective numerical error bounds for universal finite-input ceiling certification, and exponentially improved global expansions beyond the fixed-order dominant-singularity series, are not claimed here.

[write] Sources, 5 October 2026. The write read Min’s paper and Barnabei et al.’s Section 10 (arXiv version) again; the quotations in the front matter and in Section 23 are from them. Two priority points: the Catalan count of Section 3 is Guibert and Linusson’s (note there), and Min’s 2002 thesis remains uninspected (Section 23, item 4). The accepted Min–Park inputs are item 5.

Part II

Sparse fixed point crossover for alternating Baxter involutions

Bundle Report 151, archive `Baxter_Fixed_Point_Crossover_Profiles_and_Inverses_Source.zip`, delivered as `Report151.tex` (17 pages). Delivered title block “Sparse fixed point crossover for alternating Baxter involutions / Entire profiles, exact thresholds, and weighted inverses / Report 151 / 3 October 2026”. Printed as delivered, with **[write]** notes; its opening restatement of Report 149 is printed once, in Part I. Delivered Section k is Section $k + 11$ here, and its statement or equation $k.j$ is $(k + 11).j$. Section 23, which closes the report, is added in the write.

Reading conventions of Part II	Elsewhere in the report
q marks <i>pairs</i> of fixed points, x half the length; odd ascent-first objects $1 \oplus \sigma$ carry $r_m(q)$, the singleton unweighted	Part I: no weight
$E(x, q)$, $R(x, q)$, $e_m(q)$, $r_m(q)$	Part I: E , O , e_m , $o_m = r_m$ (at $q = 1$)
$\varepsilon = (-1)^m$; $T_{s,m}(w)$ with $q = w/m$, $s \in \{E, R\}$	Part I: $\varepsilon = 1/X_s$; $s \in \{e, o\}$
$y = 2x$, $H(y) = (1 + y)/\sqrt{1 - y^2}$, h_n , d_r	Part I: y a target, $H = O - 1$
$D = w d/dw$; $f(w) = (e^w - 1)/w$	Part I: $D = MP$ the discriminant
$A_{s,\ell,\varepsilon}$, $B_{s,\ell}$ profiles; $\mathcal{B}_j$ Bernoulli	Part I: $\mathcal{A}_s$, $A_s(x)$; $\mathcal{B}_n$ the class, B_j Bernoulli
$\theta_{s,m}$; inverses in the weight w at fixed m	Part I: inverses in the length
$K = \lceil \log m \rceil$, K the number of pairs; $\rho > 1$ a radius	Part I: $K_s(x)$; ρ the singularity

Abstract of this Part (as delivered). We refine the corrected alternating Baxter involution enumeration by *pairs* of fixed points. The bivariate generating functions admit positive coefficient factorizations, which lead to a sparse scaling $q = w/m$ at half-length m . At every fixed order, the normalized counts have expansions uniform on compact complex w sets, with both integer and half-integer powers of m^{-1} and an explicit parity contribution. The profiles are entire and have finite Bernoulli-polynomial algorithms. The proof combines global factorial tails with an exact parity-binomial gamma sum and a controlled growing-shift remainder. For nonnegative weights we give the exact finite-size existence and uniqueness criterion for inversion and a uniform all-orders inverse on compact positive interior windows. In particular, the limiting reverse-class threshold has no nonnegative exact inverse for all sufficiently large odd m . The associated sparse tilted laws are Poisson-type distributions, with an $O(m^{-1/2})$ total-variation rate and explicit endpoint conventions. A fresh computational companion separates exact identities from high-precision diagnostics and supplies deterministic rebuilding.

(*As delivered.*) **Scope.** This is a refinement of the corrected class intended by OEIS A347546, not a refinement of the erroneous printed recurrence. Report 149 supplies the all-length structural foundation and is included unchanged. The present report proves the fixed-point refinement and compact-window asymptotics. No exhaustive priority claim, fixed-positive-weight central limit theorem, growing-window expansion, or boundary-uniform inverse theorem is asserted.

12 The class and its fixed point convention

[write] Printed once, 5 October 2026. Report 151’s Section 1 opens with a paragraph restating Part I’s definitions (Section 1): ascent-first and reverse alternation; involutions; the Baxter condition, in its vincular

form (avoidance of 2-41-3 and 3-14-2, Part I's second, equivalent form); the coincidence of forward and inverse alternation for involutions; and the empty permutation, admitted in both even classes. It is printed once, in Part I.

Let $e_m(q)$ and $r_m(q)$ count ascent-first and reverse-first Baxter involutions of original length $2m$. An object having $2k$ fixed points contributes q^k . Thus x below marks *half-length* and q marks *pairs of fixed points*:

$$E(x, q) = \sum_{m \geq 0} e_m(q) x^m, \quad R(x, q) = \sum_{m \geq 0} r_m(q) x^m.$$

Every even involution has an even number of fixed points, since its nonfixed entries are paired in transpositions. For original odd length $2m + 1$, the ascent-first class is $1 \oplus \sigma$ with σ in the even reverse class. Its polynomial is $r_m(q)$ when q marks pairs *in addition to* the unavoidable singleton. Its actual number of fixed points is $2k + 1$. No extra factor of q is attached to that singleton. If a variable u is to mark individual fixed points, substitute $q = u^2$ in the even classes and use $uR(x, u^2)$ in the odd class.

[write] **Printed once, 5 October 2026.** Here Report 151 recalls that Report 149 [8] establishes the corrected decomposition from Min and Park's unrestricted structural results, with Ouchterlony's independent route, that the count first differs from Min's recurrence at length 20 (2168 against 2166), and that the cause is an outer block paired with its inverse. These are Part I, Sections 2–5, printed once there. “Report 149” in Part II is Part I.

Setting $q = 1$ in this report recovers the corrected counts. The association with OEIS A347546[3] refers to its intended permutation class; it does not silently identify that class with a recurrence-defined sequence that has different values.

13 Exact refinement of the structural decomposition

Put

$$C(t) = \sum_{j \geq 0} C_j t^j = \frac{1 - \sqrt{1 - 4t}}{2t}, \quad C_j = \frac{1}{j+1} \binom{2j}{j}, \quad M(x) = 1 - x^2 C(x^2) = \frac{1 + \sqrt{1 - 4x^2}}{2}. \quad (13.1)$$

The branches have constant term one, with the quotient in C interpreted by continuity at zero.

[write] **5 October 2026.** C and M are Part I's (Section 1 and (6.1)).

Theorem 13.1 (Exact bivariate system). *The counting series are the unique formal power series with constant terms one satisfying*

$$E = \frac{1 + xqR}{M}, \quad R = \frac{1}{1 - xE}. \quad (13.2)$$

Equivalently, $e_0 = r_0 = 1$ and, for $m \geq 1$,

$$e_m(q) = q r_{m-1}(q) + \sum_{j=0}^{\lfloor (m-2)/2 \rfloor} C_j e_{m-2j-2}(q), \quad (13.3)$$

$$r_m(q) = \sum_{i=0}^{m-1} e_i(q) r_{m-1-i}(q). \quad (13.4)$$

An empty sum is zero.

Proof. We recall exactly the part of the corrected decomposition that carries the statistic. An ascent-first even involution is a canonical skew sequence of primitives $1 \oplus \sigma \oplus 1$. Inversion reverses this sequence. Its outside primitives therefore occur as inverse partners, with an optional involutive central primitive. The outside position interval and the value interval to which it maps are disjoint. Hence an outside pair has *no global fixed points*, regardless of the fixed points of its standardized interior. Its generating function is the unweighted $x^2C(x^2)$: the interior is an unrestricted doubly reverse-alternating Baxter permutation, with Catalan enumeration. The center occupies identical position and value offsets, so its interior fixed points are preserved. A central primitive adds two fixed endpoints, and contributes xqR . Summing the paired sequence gives $E = (1 + xqR)/(1 - x^2C(x^2))$.

A reverse-first involution is a direct sequence of primitives $1 \ominus \tau \ominus 1$, with τ an ascent-first even involution. Inversion preserves the direct component order, so every primitive is involutive individually. Its two endpoints are exchanged and contribute no fixed points; the interior retains its statistic. Direct sums add fixed-point counts. This gives $R = (1 - xE)^{-1}$. These constructions, including their converse and uniqueness, are those established in Report 149; the present argument checks how the fixed-point statistic behaves in every component. Taking coefficients gives (13.3)–(13.4). Each right-hand coefficient has smaller half-length, proving formal uniqueness. $\square$

[write] **The case $q = 1$, 5 October 2026.** At $q = 1$, $e_m(1) = e_m$ and $r_m(1) = r_m = o_m$, and Theorem 13.1 becomes Part I's Theorem 1.1: (13.2) becomes $E = (1 + xO)/M$ and $O = 1/(1 - xE)$ (Part I, Subsection 4.3), and (13.3)–(13.4) become (1.4) and (1.3). The proof above takes the decomposition, its converse and its uniqueness from Part I and adds only how the fixed points sit in each block. At intake the bivariate recurrence agreed with the fixed-point distributions found by direct enumeration for every length $n \leq 22$; for instance the distributions at lengths 12 and 13 are $e_6(q)$ and $r_6(q)$ below.

13.1 Closed forms with positive coefficients

Since $M^2 - M + x^2 = 0$, one has

$$(M - x)^2 = M(1 - 2x).$$

Eliminating E from (13.2) gives

$$qx^2R^2 - (M - x)R + M = 0, \quad R = \frac{M}{M - x} C\left(\frac{qx^2}{1 - 2x}\right). \quad (13.5)$$

This Catalan form selects the counting branch even at $q = 0$. For $y = 2x$, define

$$H(y) = \frac{1 + y}{\sqrt{1 - y^2}}, \quad d_r = \frac{C_r}{2 \cdot 4^r} = \frac{(1/2)_r}{2(r + 1)!}, \quad h_n = [y^n]H(y). \quad (13.6)$$

The rising factorial is $(a)_r = a(a + 1) \cdots (a + r - 1)$, with $(a)_0 = 1$. Direct simplification gives

$$[q^r]R(y/2, q) = d_r y^{2r} (1 - y)^{-r} (H(y) + 1), \quad (13.7)$$

$$[q^{r+1}]E(y/2, q) = d_r y^{2r} (1 - y)^{-r} (H(y) - 1), \quad (13.8)$$

$$[q^0]E(y/2, q) = C(y^2/4). \quad (13.9)$$

These formulas hold for every $r \geq 0$. Moreover,

$$h_{2j} = h_{2j+1} = \frac{1}{4^j} \binom{2j}{j} \quad (j \geq 0). \quad (13.10)$$

Thus the fixed-weight factorizations have nonnegative coefficients, including $H - 1$, whose constant coefficient is zero. They give

$$E(x, q) = C(x^2) + qR(x, q) - qC\left(\frac{qx^2}{1-2x}\right). \quad (13.11)$$

In particular,

$$[q]e_m(q) = 2^{m-1}h_m > 0 \quad (m \geq 1), \quad [q]r_m(q) > 0 \quad (m \geq 2).$$

The degrees are $\lfloor (m+1)/2 \rfloor$ for e_m , $m \geq 1$, and $\lfloor m/2 \rfloor$ for r_m , $m \geq 2$. The exceptional reverse polynomials are $r_0 = r_1 = 1$. These statements follow by reading the smallest degrees of $H - 1$ and $H + 1$ in the factorizations.

For a small reproducible example, at half-length $m = 6$,

$$e_6(q) = 5 + 10q + 19q^2 + 10q^3, \quad r_6(q) = 10 + 35q + 34q^2 + 5q^3.$$

At $q = 1$ their totals are 44 and 84, corresponding to ascent-first original lengths 12 and 13, respectively.

14 The uniform sparse crossover theorem

Set $\varepsilon = (-1)^m$, and for $m \geq 1$ define

$$T_{R,m}(w) = 2^{-m}m^{1/2}r_m(w/m), \quad T_{E,m}(w) = 2^{-m}m^{3/2}e_m(w/m). \quad (14.1)$$

The powers of m differ because the even ascent class has an extra factor of q in its positive-weight terms. Throughout the analytic statements, w may be complex. Probability and inverse statements will explicitly restrict it to real nonnegative values.

Theorem 14.1 (All fixed orders on compact complex windows). *There are entire profiles $A_{s,\ell,\varepsilon}(w)$ and $B_{s,\ell}(w)$, for $s \in \{E, R\}$, $\ell \geq 0$ and $\varepsilon \in \{-1, 1\}$, such that for every finite $W \geq 0$ and fixed integer $L \geq 0$,*

$$T_{s,m}(w) = \sum_{\ell=0}^L \frac{A_{s,\ell,\varepsilon}(w)}{m^\ell} + \sum_{\ell=0}^L \frac{B_{s,\ell}(w)}{m^{\ell+1/2}} + O_{W,L}(m^{-L-1}), \quad |w| \leq W. \quad (14.2)$$

The error is uniform over the full complex disk. After any fixed number of w derivatives the same assertion holds, with a possibly different constant. The parity is always the actual $\varepsilon = (-1)^m$.

This is an all-*fixed*-orders expansion. Neither convergence as $L \rightarrow \infty$ nor a bound uniform in growing W or growing L follows from it. It retains a half-power profile of size $m^{-L-1/2}$ while bounding the remaining error by the next integer order m^{-L-1} .

Define the entire function and Euler operator

$$f(w) = \frac{e^w - 1}{w} = \sum_{r \geq 0} \frac{w^r}{(r+1)!}, \quad f(0) = 1, \quad D = w \frac{d}{dw}. \quad (14.3)$$

The first profiles are

$$A_{R,0,\varepsilon}(w) = \frac{f(w)}{\sqrt{2\pi}}, \quad A_{E,0,\varepsilon}(w) = \frac{e^w + 1 + 2\varepsilon}{\sqrt{2\pi}}, \quad (14.4)$$

$$B_{R,0}(w) = \frac{w}{8} {}_1F_1(3/2; 3; w), \quad B_{E,0}(w) = -wB_{R,0}(w), \quad (14.5)$$

$$A_{R,1,\varepsilon}(w) = \frac{(9 - 6w)e^w - 9f(w) - \varepsilon}{4\sqrt{2\pi}}, \quad (14.6)$$

$$A_{E,1,\varepsilon}(w) = wA_{R,1,\varepsilon}(w) - \frac{9(1 + \varepsilon)}{2\sqrt{2\pi}}, \quad (14.7)$$

$$B_{R,1}(w) = -\frac{3}{2}D(D-1)B_{R,0}(w) = -\frac{3}{2}w^2B''_{R,0}(w), \quad B_{E,1}(w) = -wB_{R,1}(w). \quad (14.8)$$

Here ${}_1F_1(a; b; w) = \sum_{n \geq 0} (a)_n w^n / ((b)_n n!)$. Keeping the terms through A_1 and B_1 leaves $O_W(m^{-2})$, uniformly on the complex disk. The leading reverse profile is parity independent, but parity returns at order m^{-1} . The ascent leading profile already distinguishes even and odd half-length, equivalently original even lengths divisible by four and congruent to two modulo four.

15 Finite algorithms for every profile

15.1 Gamma ratio coefficients and binomial moments

Let $\mathcal{B}_j(t)$ denote the Bernoulli polynomial, to distinguish it from the half-power profile $B_{s,\ell}$. Define polynomials $G_\ell(a, b)$ by

$$G_0(a, b) = 1, \quad L_h(a, b) = \frac{(-1)^{h+1}(\mathcal{B}_{h+1}(a) - \mathcal{B}_{h+1}(b))}{h(h+1)}, \quad (15.1)$$

$$\ell G_\ell(a, b) = \sum_{h=1}^{\ell} h L_h(a, b) G_{\ell-h}(a, b). \quad (15.2)$$

They are the coefficients of

$$\frac{\Gamma(z+a)}{\Gamma(z+b)} \sim z^{a-b} \sum_{\ell \geq 0} G_\ell(a, b) z^{-\ell}.$$

Only finitely many polynomial operations are required for each ℓ . The positive-real fixed-shift expansion and its remainder are classical; we use the forms in DLMF Section 5.11[9]. The moving-shift uniformity required below is proved separately in Section 17.

[write] 5 October 2026. These $G_\ell(a, b)$ are Part I's $G_r(a, b)$ of (7.7): (15.2) is the recursion for the coefficients of $\exp(\sum_h L_h(a, b) z^h)$, and L_h is the coefficient in Part I's exponent.

For an integer $r \geq 0$, let J_r be binomial with parameters $r+1$ and $1/2$, and set

$$P_\ell(r) = \mathbb{E} \left[2^\ell G_\ell \left(\frac{1 - J_r}{2}, 1 - r - \frac{J_r}{2} \right) \right], \quad (15.3)$$

$$Q_\ell(w) = \sum_{r=0}^{\ell-1} \frac{w^r}{(r+1)!} \mathbb{E} \left[(-1)^{J_r} 2^\ell G_\ell \left(\frac{1 - J_r}{2}, 1 - r - \frac{J_r}{2} \right) \right]. \quad (15.4)$$

The expectations are simply finite binomial sums; they introduce no randomness into the theorem. Put $Q_0 = 0$. The expression in J_r has degree at most ℓ : in (15.1), the two Bernoulli polynomials have a common shift $-J_r/2$, and their leading powers cancel. The recursion preserves this degree bound. Its coefficients are polynomials in r . Binomial moments are polynomials in r , since

$$\mathbb{E} J_r^k = \frac{(r+1)^k}{2^k}.$$

Consequently P_ℓ is a polynomial in r , of degree at most 2ℓ . Alternating binomial sums of a polynomial of degree ℓ vanish when $r+1 > \ell$. This explains the finite upper limit in (15.4); it is the exact contribution from the negative endpoint.

The integer profiles are

$$A_{R,\ell,\varepsilon}(w) = \frac{P_\ell(D)f(w) + \varepsilon Q_\ell(w)}{\sqrt{2\pi}}. \quad (15.5)$$

Equivalently, the formula before alternating-sum cancellation is

$$\frac{1}{\sqrt{2\pi}} \sum_{r \geq 0} \frac{w^r}{(r+1)!} \mathbb{E} \left[(1 + \varepsilon(-1)^{J_r}) 2^\ell G_\ell \left(\frac{1 - J_r}{2}, 1 - r - \frac{J_r}{2} \right) \right]. \quad (15.6)$$

The first four polynomial pairs are

$$\begin{aligned} P_0(r) &= 1, & Q_0(w) &= 0, \\ P_1(r) &= -\frac{3}{4}r(2r-1), & Q_1(w) &= -\frac{1}{4}, \\ P_2(r) &= \frac{(2r-3)(2r-1)(27r^2-r+1)}{96}, & Q_2(w) &= -\frac{w}{32}, \\ P_3(r) &= -\frac{r(2r-5)(2r-3)(2r-1)(9r^2-r+1)}{128}, & Q_3(w) &= -\frac{w^2+9w-5}{128}. \end{aligned}$$

15.2 Half powers and the ascent profiles

For a finite list $\mathbf{v}$, write $\mathfrak{e}_\ell(\mathbf{v})$ for its elementary symmetric polynomial, with $\mathfrak{e}_0 = 1$ and $\mathfrak{e}_\ell = 0$ when ℓ exceeds the list length. Define

$$B_{R,\ell}(w) = \sum_{r \geq 1} \frac{d_r w^r}{(r-1)!} (-1)^\ell \mathfrak{e}_\ell(r+1, r+2, \dots, 2r-1). \quad (15.7)$$

The list is empty at $r = 1$. Its signed symmetric polynomial is a polynomial $S_\ell(r)$ of degree at most 2ℓ , obtainable by Newton identities and polynomial power sums. Thus $B_{R,\ell} = S_\ell(D)B_{R,0}$. For example,

$$\begin{aligned} S_0(r) &= 1, \\ S_1(r) &= -\frac{3}{2}r(r-1), \\ S_2(r) &= \frac{r(r-2)(r-1)(27r-1)}{24}, \\ S_3(r) &= -\frac{r^2(r-3)(r-2)(r-1)(9r-1)}{16}, \\ S_4(r) &= \frac{r(r-4)(r-3)(r-2)(r-1)(1215r^3 - 270r^2 + 5r + 2)}{5760}. \end{aligned}$$

The leading hypergeometric expression in (14.5) follows by putting $r = n + 1$ in (15.7) for $\ell = 0$.

Set

$$c_\ell = 2^\ell G_\ell(1/2, 2), \quad (c_0, c_1, c_2, c_3, c_4) = \left(1, -\frac{9}{4}, \frac{145}{32}, -\frac{1155}{128}, \frac{36939}{2048}\right). \quad (15.8)$$

Then every ascent profile is

$$A_{E,\ell,\varepsilon}(w) = \frac{2(1+\varepsilon)c_\ell}{\sqrt{2\pi}} + wA_{R,\ell,\varepsilon}(w), \quad B_{E,\ell}(w) = -wB_{R,\ell}(w). \quad (15.9)$$

The negative sign in the latter formula is essential. In (13.11), the subtracted Catalan expression equals twice the rational contribution to qR , coefficientwise away from the constant coefficient. Its half-power contribution therefore changes $+wB_R$ to $-wB_R$.

Every series above converges locally uniformly in w : for fixed ℓ its coefficients are bounded by a polynomial in r times $1/(r+1)!$ or $1/(r-1)!$, and $0 < d_r \leq 1/2$. This proves that all profiles are entire, independently of their operator representation. Equations (15.2), (15.3), (15.4), (15.7), and (15.9) form an explicit general coefficient algorithm. A finite number of Bernoulli and symmetric polynomial calculations yields any prescribed order; evaluation of the resulting entire functions can use their convergent series.

16 Global factorial tails

For $N \geq 0$ and $\alpha > 0$ write

$$b_N(\alpha) = [y^N](1-y)^{-\alpha} = \frac{\Gamma(N+\alpha)}{\Gamma(\alpha)\Gamma(N+1)}.$$

We begin with coefficientwise estimates valid even when r is near its largest supported value. This prevents a local singular expansion from being used outside its justified range.

Lemma 16.1. *Coefficientwise,*

$$0 \leq H(y) \leq 2(1-y)^{-1/2}, \quad b_N(1/2) \leq \frac{1}{\sqrt{N+1}}. \quad (16.1)$$

For $N = m - 2r \geq 0$,

$$b_N(r+1/2) \leq \frac{m^r}{(1/2)_r \sqrt{N+1}}. \quad (16.2)$$

Proof. Use (13.10). The ratio $h_n/b_n(1/2)$ increases from an even index to the next odd index, and its odd subsequence decreases because

$$\frac{h_{2j+3}/b_{2j+3}(1/2)}{h_{2j+1}/b_{2j+1}(1/2)} = \frac{2j+1}{2j+2} \frac{(4j+4)(4j+6)}{(4j+3)(4j+5)} < 1.$$

It starts at $h_1/b_1(1/2) = 2$; the zeroth coefficient is also bounded. This proves the first inequality. For the second, induction uses $b_{N+1}(1/2)/b_N(1/2) = (2N+1)/(2N+2)$ and

$$\left(\frac{2N+1}{2N+2}\right)^2 \leq \frac{N+1}{N+2}.$$

Finally,

$$b_N(r + 1/2) = b_N(1/2) \frac{(N + 1/2)_r}{(1/2)_r}.$$

Every factor of $(N + 1/2)_r$ is at most m when $N = m - 2r \geq 0$. This gives (16.2), including $r = 0$. $\square$

By (13.7), the normalized absolute contribution of its H part at $|w| \leq W$ is bounded by

$$\sqrt{\frac{m}{m - 2r + 1}} \frac{W^r}{(r + 1)!}, \quad (16.3)$$

since $2d_r/(1/2)_r = 1/(r + 1)!$. The rational part, with H replaced by 1, is coefficientwise at most the H part. Thus twice (16.3) bounds the entire reverse contribution. For $r \leq m/4$ its square-root factor is at most $\sqrt{2}$; for all $r \leq m/2$ it is at most $\sqrt{m}$. No assertion that $m - 2r$ is comparable to m is needed at the edge.

Put $K = \lceil \log m \rceil$. For every fixed $A > 0$ and finite W ,

$$\sum_{K < r \leq m/2} |\text{normalized reverse contribution}| = o_{W,A}(m^{-A}). \quad (16.4)$$

Indeed split at $m/4$. The first factorial tail has logarithm $-K \log K + O_W(K)$, and the second starts at order m even after the factor $\sqrt{m}$. Both beat every fixed power of m^{-1} . Multiplication by any fixed polynomial in r leaves this conclusion unchanged. For $W = 0$ it is immediate. Taking absolute values is what makes the argument valid on a complex disk, rather than only on a nonnegative interval.

The ascent positive-weight terms in (13.8) have at most W times the corresponding reverse H bound, after the different normalization in (14.1). Their q^0 Catalan term is separate. Thus the same factorial-tail conclusion holds directly for the ascent class, without needing to rely on cancellation in (13.11).

17 The exact parity sum and controlled growing shifts

The low- r range is treated by an exact identity retaining both endpoints $y = 1$ and $y = -1$:

$$H(y)(1 - y)^{-r} = (1 + y)^{r+1}(1 - y^2)^{-r-1/2}. \quad (17.1)$$

Its coefficient formula is

$$[y^{m-2r}]H(y)(1 - y)^{-r} = \frac{1}{\Gamma(r + 1/2)} \sum_{\substack{0 \leq j \leq r+1 \\ m-j \text{ even} \\ m-2r-j \geq 0}} \binom{r+1}{j} \frac{\Gamma(m/2 + (1-j)/2)}{\Gamma(m/2 + 1 - r - j/2)}. \quad (17.2)$$

Both restrictions in the sum matter at finite size. For $r \leq K$, all the support restrictions $m - 2r - j \geq 0$ hold once $m \geq 3K + 1$. Only the parity restriction then remains.

Lemma 17.1 (Growing-shift remainder). *Put $z = m/2$, $a = (1 - j)/2$ and $b = 1 - r - j/2$. For each fixed $L \geq 0$, uniformly for $0 \leq r \leq \lceil \log m \rceil$ and $0 \leq j \leq r + 1$,*

$$\frac{\Gamma(z + a)}{\Gamma(z + b)} = z^{r-1/2} \left\{ \sum_{\ell=0}^L G_\ell(a, b) z^{-\ell} + O_L((1 + r)^{2L+2} z^{-L-1}) \right\}. \quad (17.3)$$

Proof. Set $t = z + (1 - j)/2$. Exactly,

$$\frac{\Gamma(z + a)}{\Gamma(z + b)} = \frac{\Gamma(t)}{\Gamma(t + 1/2)} \prod_{i=1}^r (t + 1/2 - i).$$

The ratio with fixed shift $1/2$ has, for positive real t , its usual fixed-order expansion with remainder $O_L(t^{-L-3/2})$. This follows from Stirling's formula with remainder[9]. Here $t/z = 1 + O((r+1)/z)$ is uniformly bounded away from zero. Expanding the finitely many powers of t/z gives, after removing $z^{-1/2}$,

$$U := z^{1/2} \frac{\Gamma(t)}{\Gamma(t + 1/2)} = \sum_{a'=0}^L u_{a'}(j) z^{-a'} + O_L((r+1)^{L+1} z^{-L-1}), \quad |u_{a'}(j)| \leq C_L(r+1)^{a'}.$$

For the product divided by z^r , write

$$V = \prod_{i=1}^r \left(1 + \frac{1 - j/2 - i}{z} \right).$$

The sum of the absolute shifts is $O((r+1)^2)$. Elementary-symmetric bounds therefore give

$$V = \sum_{b'=0}^L v_{b'}(r, j) z^{-b'} + O_L(e^{C(r+1)^2/z} (r+1)^{2L+2} z^{-L-1}),$$

$$|v_{b'}(r, j)| \leq C_L(r+1)^{2b'}.$$

For example, the absolute sum of degrees above L is bounded by $e^v v^{L+1}/(L+1)!$, where $v = C(r+1)^2/z$. This v tends to zero uniformly in the stated range. On multiplying the two expansions, omitted mixed degrees $a' + b' \geq L+1$ are bounded by the same remainder: their polynomial degree is at most $a' + 2b' \leq 2(a' + b')$, and each additional inverse power of z is accompanied by the bounded ratio $(r+1)^2/z$. The remainders of U and V obey the same bound after multiplication.

This proves a polynomially controlled expansion of UV . For fixed r, j , uniqueness of asymptotic coefficients identifies it with the $G_\ell(a, b)$ from (15.2). Hence (17.3) holds. In particular, no fixed-shift theorem has been applied directly with uncontrolled growing parameters. $\square$

17.1 Completion of the uniform proof

Multiply (17.2) and (17.3) by $d_r(w/m)^r \sqrt{m}$. The exact prefactor is

$$\frac{d_r 2^{1/2-r}}{\Gamma(r+1/2)} = \frac{1}{\sqrt{2\pi} 2^r (r+1)!}.$$

Since $\mathbf{1}_{m-j \text{ even}} = (1 + \varepsilon(-1)^j)/2$, the binomial sum becomes the expectation in (15.6). This checks the normalization, including the factor of two. In the absolute remainder estimate the sum of the binomial coefficients is 2^{r+1} , which cancels 2^r up to a constant. Summing the low- r errors is bounded by

$$C_L m^{-L-1} \sum_{r \leq K} \frac{(1+r)^{2L+2} W^r}{(r+1)!} = O_{W,L}(m^{-L-1}). \quad (17.4)$$

Extending the truncated profile sums to all r costs only another superpolynomial factorial tail, by Section 16.

For the rational part and $r \geq 1$ the exact coefficient is

$$[y^{m-2r}](1-y)^{-r} = \binom{m-r-1}{r-1} = \frac{m^{r-1}}{(r-1)!} \prod_{h=r+1}^{2r-1} \left(1 - \frac{h}{m}\right), \quad m \geq 2r. \quad (17.5)$$

Expanding this finite product in elementary symmetric polynomials gives (15.7). In the low range the omitted degrees have the same exponential elementary-symmetric bound used in Lemma 17.1. After normalization their error is in fact $O_{W,L}(m^{-L-3/2})$. The rational $r = 0$ coefficient is zero for $m \geq 1$. The already controlled global tails handle all larger r . This proves (14.2) for R .

The zero-weight ascent term vanishes at odd m . At even m use

$$C_{m/2} = \frac{2^m \Gamma(m/2 + 1/2)}{\sqrt{\pi} \Gamma(m/2 + 2)}$$

and the fixed-shift gamma expansion. Its normalized coefficients are $2(1 + \varepsilon)c_\ell/\sqrt{2\pi}$. Now (13.11), or the direct $H - 1$ factorization, gives (15.9) and the same uniform remainder for E . Finally apply Cauchy's derivative estimate on a larger complex disk to the analytic remainder. This proves all the derivative claims and completes Theorem 14.1.

18 Exact inverse existence and finite thresholds

We now take $w \geq 0$ real and ask for the weight $q = w/m$ that produces a prescribed normalized count Y . This is an inversion in the weight, not an inversion in the original permutation length.

Theorem 18.1 (Exact nonnegative inverse). *For E with $m \geq 1$, or R with $m \geq 2$, the polynomial $T_{s,m}(w)$ is strictly increasing on $[0, \infty)$ and maps it onto $[\theta_{s,m}, \infty)$, where*

$$\theta_{E,m} = \begin{cases} m^{3/2} C_{m/2} / 2^m, & m \text{ even,} \\ 0, & m \text{ odd,} \end{cases} \quad \theta_{R,m} = \frac{\sqrt{m} h_m}{2}. \quad (18.1)$$

Thus $T_{s,m}(w) = Y$ has a unique positive solution exactly when $Y > \theta_{s,m}$; equality has the unique nonnegative solution $w = 0$; and a smaller target has no nonnegative solution.

Proof. The exact coefficient factorizations give nonnegative coefficients and a strictly positive linear coefficient in the indicated ranges. The derivative is therefore strictly positive for every $w \geq 0$, and the polynomial tends to infinity. Its value at zero is (18.1), by (13.9) and (13.7). $\square$

For the excluded $r_0 = r_1 = 1$, the normalized expression, when defined, is constant in w , so the nonconstant-polynomial conclusion must not be applied.

18.1 Why a limiting endpoint is not an exact threshold

At $w = 0$, (14.6) and the vanishing of all reverse half-power profiles give

$$\theta_{R,m} = \frac{1}{\sqrt{2\pi}} \left(1 - \frac{\varepsilon}{4m} + O(m^{-2})\right). \quad (18.2)$$

Consequently the fixed target $Y = 1/\sqrt{2\pi}$ has a positive exact solution for all sufficiently large even m , but *no nonnegative exact solution* for all sufficiently large odd m . A model root at $w = 0$ is

therefore not a finite-size existence statement. No rounding of Y or replacement of $\theta_{R,m}$ by its limit is permitted in Theorem 18.1.

For ascent at sufficiently large even m , the finite threshold is below its limiting value $4/\sqrt{2\pi}$; at odd m it is exactly zero. For example, the fixed-shift expansion gives the former threshold $(4/\sqrt{2\pi})(1 - 9/(4m) + O(m^{-2}))$. The exact formula, not this truncation, is the criterion for a finite input. The all-orders inverse theorem below deliberately remains on positive interior windows.

18.2 The two even boundary roots

The exact limiting targets themselves admit a more specific result, separate from uniform inversion over a moving family of targets.

Proposition 18.2 (Threshold-specific boundary expansion). *For all sufficiently large even m , let $w_{E,m}$ and $w_{R,m}$ be the positive roots at targets $4/\sqrt{2\pi}$ and $1/\sqrt{2\pi}$, respectively. Then*

$$w_{E,m} = \frac{9}{m} - \frac{451}{8m^2} + \frac{81\sqrt{2\pi}}{8m^{5/2}} + O(m^{-3}), \quad (18.3)$$

$$w_{R,m} = \frac{1}{2m} - \frac{\sqrt{2\pi}}{8m^{3/2}} + \frac{\pi/16 + 11/48}{m^2} + O(m^{-5/2}). \quad (18.4)$$

The weights $q = w/m$ consequently have order m^{-2} at these two targets. For odd m , the reverse target has no nonnegative root for all sufficiently large m , while the ascent limiting target zero has only the exact root $w = 0$.

Proof. The finite threshold expansions and exact strict monotonicity give existence. Evaluating the compact-uniform profiles at zero and at a sufficiently large constant multiple of $1/m$ localizes each even root to $w = O(m^{-1})$. Write $S = \sqrt{2\pi}$. For even ascent, expansion of the entire profiles in this shrinking interval gives

$$S(T_{E,m}(w) - 4/S) = w + \frac{w^2}{2} - \frac{9}{m} - \frac{w}{4m} + \frac{145}{8m^2} - \frac{Sw^2}{8\sqrt{m}} + O(m^{-3}). \quad (18.5)$$

Substitution of $w = a/m + b/m^2 + c/m^{5/2}$ yields successively $a = 9$, $b = -451/8$, and $c = 81S/8$. There is no $m^{-3/2}$ term. The derivative of the left side tends uniformly to one. The mean value theorem converts the $O(m^{-3})$ residual into that root error.

For even reverse the corresponding expansion is

$$S(T_{R,m}(w) - 1/S) = \frac{w}{2} + \frac{w^2}{6} - \frac{1}{4m} - \frac{3w}{8m} + \frac{1}{32m^2} + \frac{Sw}{8\sqrt{m}} + O(m^{-5/2}). \quad (18.6)$$

Inserting $w = a/m + b/m^{3/2} + c/m^2$ gives $a = 1/2$, $b = -S/8$ and $c = \pi/16 + 11/48$. The derivative tends to $1/2$, so residual-to-root conversion gives (18.4). The shrinking-interval remainders in both calculations follow from Theorem 14.1 at fixed order on a fixed disk, followed by Taylor expansion of its entire profiles. The odd statements are the exact threshold conclusions already proved. $\square$

These are expansions for two specified targets, not a theorem uniform over arbitrary targets approaching the threshold. In particular they do not remove the odd reverse obstruction.

[write] **Numerical check, 5 October 2026.** At intake the exact polynomials were evaluated to $m = 321$ (60-digit arithmetic). The scaled errors $(w_{E,m} - \text{prediction})m^3$ of (18.3) are 320.1, 360.7, 387.9, 405.4

and $(w_{R,m} - \text{prediction})m^{5/2}$ of (18.4) are $-0.342, -0.377, -0.403, -0.423$ at $m = 40, 80, 160, 320$; their increments shrink by factors of about 0.65 and 0.76, consistent with the stated orders but, for $w_{E,m}$, with a large constant (near -450 if the increments keep shrinking so), so the expansion is slow to settle at moderate m . Uncertified; see Section 23, item 1.

19 Uniform inverse expansions in the positive interior

Let $g_s = A_{s,0,\varepsilon}$ and $b_s = B_{s,0}$, with $\varepsilon = (-1)^m$. Both leading functions are strictly increasing on $[0, \infty)$: $g'_E(w) = e^w/\sqrt{2\pi}$, while

$$g'_R(w) = \frac{1}{\sqrt{2\pi}} \int_0^1 t e^{tw} dt > 0, \quad g'_R(0) = \frac{1}{2\sqrt{2\pi}}.$$

The integral follows from $f(w) = \int_0^1 e^{tw} dt$.

Theorem 19.1 (All fixed orders of interior inversion). *Fix $0 < a \leq b < \infty$. For targets $Y = g_s(w_0)$ with $a \leq w_0 \leq b$, the unique positive exact root w_m exists for all sufficiently large m , uniformly in w_0 . For every fixed integer $N \geq 0$ it has an expansion*

$$w_m = w_0 + \sum_{j=1}^N u_j(w_0, \varepsilon) m^{-j/2} + O_{a,b,N}(m^{-(N+1)/2}). \quad (19.1)$$

The coefficients are given by finite formal series reversion using the profiles of Theorem 14.1.

Proof. Choose a slightly larger compact positive interval containing $[a, b]$ in its interior. The derivative of g_s has a positive minimum there. Uniform convergence of $T_{s,m}$ and its derivative first places the exact root close to w_0 and bounds its derivative away from zero, uniformly. Existence also follows from the strict gap to the limiting zero-weight threshold and Theorem 18.1.

Put $t = m^{-1/2}$ and define

$$F_{2\ell}(w) = A_{s,\ell,\varepsilon}(w), \quad F_{2\ell+1}(w) = B_{s,\ell}(w).$$

Inserting a formal series $w = w_0 + \sum_{j \geq 1} u_j t^j$ into $\sum_{k \geq 0} F_k(w) t^k = F_0(w_0)$ determines each next coefficient linearly, with coefficient $F'_0(w_0) = g'_s(w_0) > 0$. The analytic expansion and its derivative bounds validate any fixed truncation uniformly. The residual after N coefficients is $O(t^{N+1})$. The mean value theorem and the uniform positive derivative bound turn this residual into the same order of root error, proving (19.1). $\square$

For a concrete finite recursion, if $v_{n-1}(t) = w_0 + \sum_{j=1}^{n-1} u_j t^j$, then

$$u_n = - \frac{[t^n] \sum_{k=0}^n F_k(v_{n-1}(t)) t^k}{g'_s(w_0)}. \quad (19.2)$$

Taylor expansion to degree n makes every step finite. In particular,

$$u_1 = - \frac{b_s(w_0)}{g'_s(w_0)}, \quad (19.3)$$

$$u_2 = - \frac{A_{s,1,\varepsilon}(w_0) + b'_s(w_0)u_1 + \frac{1}{2}g''_s(w_0)u_1^2}{g'_s(w_0)}. \quad (19.4)$$

For $w_0 > 0$, $b_R(w_0) > 0$, so $u_1 < 0$ in the reverse class; $b_E(w_0) = -w_0 b_R(w_0) < 0$, so $u_1 > 0$ in the ascent class. The actual weight satisfies $q_m = w_m/m$ and is obtained by dividing (19.1) by m .

19.1 Explicit leading inverses and their domains

For ascent,

$$w_0 = \log(\sqrt{2\pi}Y - 1 - 2\varepsilon). \quad (19.5)$$

This gives a positive root for $Y > 4/\sqrt{2\pi}$ if $\varepsilon = 1$, and $Y > 0$ if $\varepsilon = -1$. For reverse, put $h = \sqrt{2\pi}Y > 1$. The positive root is

$$w_0 = -W_{-1}\left(-\frac{e^{-1/h}}{h}\right) - \frac{1}{h}. \quad (19.6)$$

Indeed $f(w_0) = h$ becomes $e^{w_0} = 1 + hw_0$. Its Lambert argument lies in $(-1/e, 0)$. The principal branch W_0 equals $-1/h$ and produces the extraneous zero root of this multiplied equation; the branch W_{-1} produces the unique positive root. Strict increase of f confirms the branch and excludes any other positive solution. The exact threshold test in Section 18 still governs every finite m . The compact-interior theorem makes no statement uniform as $a \downarrow 0$ or for target sequences approaching an endpoint at an unspecified rate.

[write] **The transseries volume, 5 October 2026.** (19.6) is an instance of the Lambert core `p0:thm:lambert-core` (3) of the transseries volume [13] after the change of variable $X = w_0 + 1/h$. Indeed $e^{w_0} = 1 + hw_0 = hX$ and $w_0 = X - 1/h$ give $X - \log X = L$ with $L = \log h + 1/h$: the core with $a = 1$, $b = -1$. Since $h > 1$, $L > 1 = L_c$, so there are exactly two positive solutions; the large one is $X = -W_{-1}(z)$ with $z = -e^{-L} = -e^{-1/h}/h$, which is (19.6), and the small one is $X = -W_0(z) = 1/h$, the extraneous root $w_0 = 0$ named above. The ascent inverse (19.5) is a logarithm, without Lambert W . Theorem 19.1 and Proposition 18.2 are reversions of a simple root in powers of $m^{-1/2}$; they resemble the volume's perturbed inversion `p0:thm:perturbed-inversion` but are not instances of it, the perturbation being an asymptotic expansion rather than an analytic family.

20 Sparse tilted fixed point laws

For a real $w > 0$, assign weight $(w/m)^K$ to an object with K fixed-point pairs and normalize within its class. Write $\mathcal{L}_{s,m,w}$ for the resulting law of K . Its probability generating function is

$$\mathcal{P}_{s,m,w}(z) = \frac{T_{s,m}(wz)}{T_{s,m}(w)}. \quad (20.1)$$

For ascent with odd m , the zero-weight total is zero, so the original measure at $w = 0$ is undefined. We will use only its specified continuous extension there. This distinction is necessary even though a limiting probability generating function exists.

Theorem 20.1 (Sparse Poisson-type laws). *For every fixed finite W , the following probability laws satisfy*

$$\sup_{0 \leq w \leq W} d_{\text{TV}}(\mathcal{L}_{s,m,w}, \mathcal{L}_{s,\varepsilon,w}) = O_W(m^{-1/2}), \quad (20.2)$$

where for odd ascent the finite- m law at zero is interpreted by continuous extension. The limiting probability generating functions are

$$\mathcal{P}_{E,+,w}(z) = \frac{e^{wz} + 3}{e^w + 3}, \quad (20.3)$$

$$\mathcal{P}_{E,-,w}(z) = \frac{e^{wz} - 1}{e^w - 1}, \quad (20.4)$$

$$\mathcal{P}_{R,w}(z) = \frac{f(wz)}{f(w)} = \frac{e^{wz} - 1}{z(e^w - 1)}. \quad (20.5)$$

Apparent singularities are removed by continuity. At $w = 0$ these laws are, respectively, δ_0 , δ_1 , and δ_0 . The same $O(m^{-1/2})$ conclusion holds for each fixed polynomial moment and each fixed exponential moment, uniformly on the stated w window. The probability generating functions themselves have all fixed asymptotic orders, obtained by series division.

Proof. For ascent with even m and for reverse, the leading denominators are bounded away from zero on $[0, W]$. The uniform complex theorem applied with wz on a bounded disk, followed by division, gives (20.3) and (20.5), with error $O_{W,\rho}(m^{-1/2})$ on every fixed circle $|z| = \rho > 1$.

For odd ascent, the exact polynomial $T_{E,m}(w)$ has a factor w and a positive linear coefficient. All its profiles also vanish at zero. Divide the exact polynomial and the expansion by this factor. The divided analytic remainder keeps the same uniform order on a smaller disk, by Cauchy's formula or the maximum principle on a larger circle. Its leading divided denominator is $f(w)/\sqrt{2\pi}$, bounded away from zero on $[0, W]$. In the numerator of the generating function the factor contributes the necessary additional z :

$$\mathcal{P}_{E,m,w}(z) = z \frac{T_{E,m}(wz)/(wz)}{T_{E,m}(w)/w}.$$

This proves (20.4), including its continuous extension z at $w = 0$. The finite- m extension is also z , since the smallest supported number of pairs is one. It does not define the original zero-weight normalization, which remains $0/0$.

On $|z| = \rho > 1$, Cauchy's coefficient estimate bounds each probability coefficient difference by $C_{W,\rho} m^{-1/2} \rho^{-k}$. Summing over $k \geq 0$ proves the total-variation bound (one half of the absolute coefficient sum). Taking ρ larger also makes $\sum k^p \rho^{-k}$ or $\sum e^{\lambda k} \rho^{-k}$ finite for any fixed nonnegative integer p or real λ . This proves the moment claims. At every fixed order the same division argument applies to Theorem 14.1. $\square$

The distributions have simple interpretations. In (20.3) there is a $\text{Poisson}(w)$ component with weight $e^w/(e^w + 3)$ and an additional atom at zero with weight $3/(e^w + 3)$. In (20.4), K is $\text{Poisson}(w)$ conditioned to be positive. In (20.5), $K + 1$ has that positive-conditioned Poisson law. For example, their limiting means are

$$\frac{we^w}{e^w + 3}, \quad \frac{we^w}{e^w - 1}, \quad \frac{we^w}{e^w - 1} - 1,$$

with endpoint values $0, 1, 0$. The actual fixed-point counts are $2K$ in the even classes and $2K + 1$ for odd ascent-first length. These injective deterministic transformations preserve total-variation distance and transport all the conclusions above.

21 Reproducibility and verification

The source package contains the report's L^AT_EX source, PDF, newly written exact companion, tests, build instructions, and the unchanged Report 149 source and PDF as the structural foundation. It does not require that foundation's old code. The companion implements both the polynomial recurrences (13.3)–(13.4) and an independent exact evaluation of (13.7)–(13.9). It also implements the finite profile coefficient formulas and checks the printed tables against them. The test suite uses explicit failures that remain active under both ordinary and optimized Python.

Exact checks compare supported coefficient polynomials, parity-binomial coefficients with their finite support, rational products, profile identities, and finite endpoint criteria. High-precision complex-window and inverse-residual calculations are labeled *diagnostics*. A finite set of numerical

samples cannot prove the uniform analytic bounds in Theorem 14.1; those bounds are established by the factorial-tail and growing-shift arguments in Sections 16 and 17. Likewise numerical inversions do not establish the existence theorem, whose proof uses exact nonnegative coefficients.

The deterministic build uses fixed metadata and a fixed source epoch, disables $\text{\TeX}$ shell escape, and creates new output destinations. The builder refuses overwriting and symlink paths. Its source ZIP uses an explicit file allowlist, sorted members, fixed timestamps, and recorded checksums. A fresh replay can therefore test exact arithmetic, optimized-mode behavior, PDF regeneration, and package identity separately. The detailed commands and the exact tested ranges are given in the companion’s README and executable tests. Rebuilding a PDF byte-for-byte requires the same $\text{\TeX}$ toolchain and fonts; mathematical reproducibility does not depend on identical rendering software.

[write] **The companion here, 5 October 2026.** Report 151’s companion is shipped as `code/151-fixedpt-companion.py`, `code/151-fixedpt-test_companion.py`, `code/151-fixedpt-build.py` and `data/151-fixedpt-requirements.txt`; its README, its PDF and the copy of Report 149 under `foundation/` are not shipped (the copy is Part I). The report README gives the rerun route from the arrival commit.

22 Literature context and limits

The all-length class foundation is the corrected decomposition of Report 149, using the unrestricted Min–Park structural statements and Ouchterlony’s independent class description. Barnabei, Bonetti, Castronuovo, and Silimbani[5] discuss the alternating 2413, 3142-avoiding involution class in their Section 10 and relate it to alternating Baxter involutions, but cite the old recurrence. That citation does not repair the outside-block restriction.

General algebraic enumeration of permutation classes with finitely many simple permutations, including involution restrictions and alternating properties, predates this work. Brignall, Huczynska, and Vatter[6] provide this machinery and the inverse-pair skew decomposition. Consequently neither general algebraicity nor the inverse-pair idea is claimed as new here. The contribution presented is a derived fixed-point refinement and an explicit compact-window asymptotic extension of the corrected enumeration.

Park and Rizzolo[10], in the September 2026 revision of their preprint, study fixed-point-biased pattern-avoiding involutions for length-three forbidden patterns and monotone forbidden patterns. Their work includes biased laws, parity phenomena, and phase transitions in those other classes. Their bias marks individual fixed points, whereas q here marks pairs. It is relevant prior context, not a source for the present class-specific profiles. Broad claims to have first introduced fixed-point bias, parity-conditioned limits, or phase transitions would therefore be unjustified.

The literature check is bounded, not an exhaustive novelty clearance. The present conclusions do not include growing compact windows, uniformity in asymptotic order, a fixed-positive- q central limit theorem, a global complex inverse, or a general moving-target boundary inverse beyond the two threshold-specific expansions in Proposition 18.2. The asymptotic remainder constants are qualitative, not certified finite-input numerical bounds. The exact finite threshold test remains available independently. Nothing in this report is a public sequence edit, an external erratum, a submission, or author correspondence.

[write] **Sources, 5 October 2026.** Park and Rizzolo’s preprint was located at intake (arXiv:2512.25006v2, revised 1 September 2026) but not read by the write; Brignall–Huczynska–Vatter was not read either. The open items of this section are Section 23, items 6–8.

23 Corrections on record, and further questions and research (both Parts)

[write] **Standing rule, 5 October 2026.** This section was added in the write. Subsection 23.1 records, with proofs, the published statements that Part I shows to be wrong; under Vladimir’s standing rule of 4 October 2026 they stay on record with a counterexample and an explicit proof of failure. Subsection 23.2 collects every claim the two manuscripts leave unproved and every limitation they state as a boundary of their results, with its source, the argument sketch it came with, and what is missing. No claim of Report 149 or Report 151 was found to be wrong.

23.1 The published recurrence and the OEIS data

Min’s notation (front matter, “The published recurrence, the OEIS entry, and the inverses”): $B_n = \mathcal{B}_n$, $RB_n = \mathcal{R}_n$, $B_n(I)$ and $RB_n(I)$ their involutions, $b_n = |B_n(I)| = a_n$. Min uses Part I’s strict four-index Baxter definition (her p. 253) and the same alternation convention.

Remark 23.1 ([write] Min (2021), Theorem 2.7, 5 October 2026). *The statement.* Min [2, p. 255] states: “If $b_n = |B_n(I)|$ then its recursive formula is

$$b_{2n-1} = \sum_{k=1}^{n-1} b_{2k-2} \cdot b_{2n-2k-1} \quad (n \geq 2), \quad b_{2n} = b_{2n-1} + \sum_{k=\lceil \frac{n}{2} \rceil}^{n-1} b_{2n-2k-1} \cdot b_{4k-2n} \quad (n \geq 3).$$

Note that we have $b_0 = 1$ by considering the empty permutation and $b_1 = b_2 = 1$, by a direct computation.” The proof adds $b_4 = 2$ (“since 1324 and 3412 are the only permutations in $B_4(I)$ ”).

The faulty step. For the even formula the proof (p. 256) writes an involution $\pi \in B_{2n}(I)$ with first entry $2k+1$, $n \leq 2k$, as $\pi = \pi_1 \pi_2 \pi_3$, where $\pi_1 = (2k+1)a_2 \cdots a_{2n-2k-1}(2n)$, $\pi_2 = a_{2n-2k+1} \cdots a_{2k}$ and $\pi_3 = a_{2k+1} \cdots a_{2n}$, and asserts: “If we consider the permutation $\pi_1^* = 1(a_2 - 2k) \cdots (a_{2n-2k-1} - 2k)(2n - 2k)$ corresponding to π_1 , then $\pi_1^* \in B_{2n-2k}(I)$ and $(a_2 - 2k - 1)(a_3 - 2k - 1) \cdots (a_{2n-2k-1} - 2k - 1) \in RB_{2n-2k-2}(I)$. So $|RB_{2n-2k-2}(I)| = |B_{2n-2k-1}(I)| = b_{2n-2k-1}$ by Lemma 2.5. Since π is an involution, $\pi_1^* = \pi_3$.” Both assertions about π_1^* are false in general. The block π_1 occupies the positions $1, \dots, 2n - 2k$ and takes the values $2k+1, \dots, 2n$; since $\pi = \pi^{-1}$, the block π_3 occupies exactly those values’ positions and takes the values $1, \dots, 2n - 2k$, and $\pi(i) = v$ holds if and only if $\pi(v) = i$. Hence the standardization of π_3 is the *inverse* of π_1^* (Part I’s $\text{std}(\pi_3) = \text{std}(\pi_1)^{-1}$, Section 5), and π_1^* need not be an involution. By Part I (4.1), $\pi_1^* = 1 \oplus \sigma \oplus 1$ with σ ranging over *all* of $RB_{2n-2k-2}$, which has C_{n-k-1} elements (3.3), and not over its $b_{2n-2k-1}$ involutions. The corrected even formula is

$$b_{2n} = b_{2n-1} + \sum_{k=\lceil n/2 \rceil}^{n-1} C_{n-k-1} b_{4k-2n} \quad (n \geq 1),$$

which is Part I’s (1.4) with $m = n$ and $j = n - k - 1$.

What is right. The odd formula, Lemma 2.5 ($|B_{2n+1}(I)| = |RB_{2n}(I)|$), the case $k = 0$, the reduction to $n \leq 2k$ (Case 1), and the count b_{4k-2n} of the central block are correct; Proposition 23.2 gives the exact extent of the error.

Proposition 23.2 ([write] The published recurrence against the class, 5 October 2026). *Let $\hat{b}_n$ be the sequence defined by Min’s Theorem 2.7 with $\hat{b}_0 = \hat{b}_1 = \hat{b}_2 = 1$ and $\hat{b}_4 = 2$ (the even formula at $n = 2$ also gives 2). Then:*

- (i) the odd formula holds for the class counts a_n for every $n \geq 2$;
(ii) the even formula holds for the class counts for $3 \leq n \leq 9$ and fails for every $n \geq 10$, where its right-hand side falls short of a_{2n} by

$$\sum_{j=4}^{\lfloor (n-2)/2 \rfloor} (C_j - o_j) e_{n-2j-2} \geq 2e_{n-10} \geq 2;$$

- (iii) $\hat{b}_n = a_n$ for $n \leq 19$ and for $n = 21$, and $\hat{b}_n < a_n$ for $n = 20$ and for every $n \geq 22$; in particular $a_{20} - \hat{b}_{20} = 2$.

Proof. Write $\hat{e}_m = \hat{b}_{2m}$ and $\hat{o}_m = \hat{b}_{2m+1}$. In the odd formula put $n = m + 1$ and $i = k - 1$: it reads $o_m = \sum_{i=0}^{m-1} e_i o_{m-1-i}$ for $m \geq 1$, which is Part I's (1.3); this proves (i). In the even formula put $n = m$ and $j = m - k - 1$. Then $2n - 2k - 1 = 2j + 1$ and $4k - 2n = 2(m - 2j - 2)$, and $\lceil m/2 \rceil \leq k \leq m - 1$ becomes $0 \leq j \leq m - 1 - \lceil m/2 \rceil = \lfloor (m - 2)/2 \rfloor$; the formula reads

$$e_m = o_{m-1} + \sum_{j=0}^{\lfloor (m-2)/2 \rfloor} o_j e_{m-2j-2}. \quad (23.1)$$

Part I's (1.4) is the same with C_j in place of o_j . Two facts about the coefficients: $o_j = r_j$ counts the involutions in $\mathcal{R}_{2j}$ (Part I, Subsection 4.2), a subset of $\mathcal{R}_{2j}$, which has C_j elements by (3.3), so $0 \leq o_j \leq C_j$; and $(o_0, \dots, o_4) = (a_1, a_3, a_5, a_7, a_9) = (1, 1, 2, 5, 12)$, so $o_j = C_j$ for $j \leq 3$ and $C_4 - o_4 = 14 - 12 = 2$. Also $o_m \geq e_0 o_{m-1} = o_{m-1}$ by (1.3), so $o_m \geq 1$, and $e_m \geq o_{m-1} \geq 1$ for $m \geq 1$, so $e_m \geq 1$ for all m .

(ii) With the class counts, e_m minus the right-hand side of (23.1) is $\sum_j (C_j - o_j) e_{m-2j-2}$, a sum of nonnegative terms. For $3 \leq m \leq 9$ every j in it is at most 3, so it vanishes. For $m \geq 10$ the term $j = 4$ occurs and equals $2e_{m-10} \geq 2$.

(iii) The sequence $\hat{b}$ satisfies, for every $m \geq 1$,

$$\hat{o}_m = \sum_{i=0}^{m-1} \hat{e}_i \hat{o}_{m-1-i}, \quad \hat{e}_m = \hat{o}_{m-1} + \sum_{j=0}^{\lfloor (m-2)/2 \rfloor} \hat{o}_j \hat{e}_{m-2j-2},$$

with $\hat{e}_0 = \hat{o}_0 = 1$: the odd formula gives the first equation as in (i); the even formula gives the second for $m \geq 3$, and for $m = 1, 2$ it gives $\hat{e}_1 = 1 = \hat{b}_2$ and $\hat{e}_2 = 2 = \hat{b}_4$, the values Min states. As for o_m and e_m , $\hat{o}_m \geq \hat{o}_{m-1}$, so $\hat{o}_m \geq 1$ and $\hat{e}_m \geq 1$.

Step 1: $\hat{e}_m \leq e_m$ and $\hat{o}_m \leq o_m$ for all m . By induction on m (each right-hand side uses smaller indices only): every term is a product of nonnegative numbers, each at most the corresponding class count, and the coefficients satisfy $\hat{o}_j \leq o_j \leq C_j$.

Step 2: $\hat{e}_m = e_m$ for $m \leq 9$ and $\hat{o}_m = o_m$ for $m \leq 10$. By induction: for $m \leq 9$ the even equation uses $\hat{o}_j$ only for $j \leq 3$, where $\hat{o}_j = o_j = C_j$, so both even equations have the same terms; the odd equation for $\hat{o}_m$ uses $\hat{e}_i$ for $i \leq m - 1 \leq 9$ only.

Step 3: $\hat{e}_m < e_m$ for $m \geq 10$. By Steps 1–2, every term of the even equation for $\hat{e}_m$ is at most the corresponding term of (1.4), and the term $j = 4$ is $\hat{o}_4 \hat{e}_{m-10} \leq 12 e_{m-10} = C_4 e_{m-10} - 2e_{m-10}$. So $\hat{e}_m \leq e_m - 2e_{m-10} \leq e_m - 2$. At $m = 10$ all other terms are equal (Step 2), so $e_{10} - \hat{e}_{10} = 2e_0 = 2$.

Step 4: $\hat{o}_m < o_m$ for $m \geq 11$. The odd equation for $\hat{o}_m$ contains the term $i = 10$, and $\hat{e}_{10} \hat{o}_{m-11} \leq (e_{10} - 2) o_{m-11} \leq e_{10} o_{m-11} - 2$; the other terms are at most their counterparts. So $\hat{o}_m \leq o_m - 2$.

In terms of n : $\hat{b}_{2m} = a_{2m}$ for $2m \leq 18$ and $\hat{b}_{2m} < a_{2m}$ for $2m \geq 20$; $\hat{b}_{2m+1} = a_{2m+1}$ for $2m+1 \leq 21$ and $\hat{b}_{2m+1} < a_{2m+1}$ for $2m+1 \geq 23$. $\square$

Remark 23.3 ([**write**] The counterexample, 5 October 2026). The two involutions $\pi^{(1)}, \pi^{(2)}$ of length 20 printed in Section 5 satisfy Min’s own definitions: each is an involution, alternates as $\pi_1 < \pi_2 > \pi_3 < \dots$ (so does its inverse, which is itself), and satisfies the strict four-index Baxter condition of her p. 253 on all $\binom{20}{4} = 4845$ quadruples (checked by the delivered companion, at intake, and again by the write). For both, $n = 10$ and the first entry is $11 = 2k + 1$ with $k = 5$, the entry 20 stands at position $2n - 2k = 10$, and the last ten entries are $1, \dots, 10$: the shape of Min’s Lemma 2.1, in her Case 2 with $n = 2k$ and empty middle block π_2 . Their blocks $\pi_1^* = \text{std}(\pi_1)$ are $(1, 9, 5, 7, 6, 8, 3, 4, 2, 10)$ and $(1, 9, 7, 8, 3, 5, 4, 6, 2, 10)$, which are not involutions (the first sends $2 \mapsto 9$ and $9 \mapsto 2$ but $3 \mapsto 5$ and $5 \mapsto 6$); for each witness $\text{std}(\pi_3)$ is the inverse of its π_1^* , as in Remark 23.1, and equals the other witness’s π_1^* . So they contradict the proof’s assertion $\pi_1^* \in B_{10}(I)$. Min’s term for $k = 5$ is $b_9 b_0 = 12$, the number of involutions π_1^* ; the involutions of this shape are exactly the $C_4 = 14$ permutations $A \oplus A^{-1}$, $A = 1 \oplus \sigma \oplus 1$, $\sigma \in RB_8$ (Part I, Section 4.1), and the two not counted are $\pi^{(1)}$ and $\pi^{(2)}$. All other terms of the even formula at $n = 10$ agree with the class, so $|B_{20}(I)| = 2168$ while Theorem 2.7 gives 2166 (Proposition 23.2; the arithmetic is $1792 + (12 + 10 + 16 + 44 + 292) = 2166$ against $1792 + (14 + 10 + 16 + 44 + 292) = 2168$, the terms listed for $k = 5, \dots, 9$). Independently of the decomposition, two brute-force enumerations written at intake count 2168 alternating Baxter involutions of length 20 (front matter).

Remark 23.4 ([**write**] Min’s list of values, 5 October 2026). Min’s paper ends (p. 257): “The first few values of b_n ($n = 0, 1, 2, 3, \dots$) are 1, 1, 1, 1, 2, 2, 3, 5, 8, 12, 16, 32, 44, 84, 105, 231, 292, 636, 768, 1792, 2166, 5080, 6012, 14592, 17234, 42198, 49336, $\dots$ ”. By Proposition 23.2 the values for $n = 20$ and $n = 22, \dots, 26$ are those of the recurrence, not of the class: the class has 2168, 6014, 14594, 17252, 42204, 49360 there (Part I’s table in Section 1 gives $n \leq 24$). The first twenty values and $b_{21} = 5080$ are right.

Remark 23.5 ([**write**] The data of OEIS A347546, 5 October 2026). The entry’s data (revision #20, read on 5 October 2026) are Min’s recurrence values $\hat{b}_n$ for $n = 0, \dots, 41$, and its Python program computes them: its lower summation limit $\text{round}(n/4)$ is one less than Min’s $\lceil n/4 \rceil$ when $n/2 \equiv 1 \pmod{4}$ (Python rounds halves to even), but the extra term then involves $b(-2)$, which the program evaluates to 0. By Proposition 23.2, 21 of the 42 terms are wrong as counts of the class the entry names, and the other 21 are right:

n	A347546	class a_n	difference	n	A347546	class a_n	difference
20	2166	2168	2	31	1066497	1066875	378
21	5080	5080	0	32	1234242	1235240	998
22	6012	6014	2	33	3164870	3166040	1170
23	14592	14594	2	34	3651296	3653440	2144
24	17234	17252	18	35	9436968	9440960	3992
25	42198	42204	6	36	10866726	10874496	7770
26	49336	49360	24	37	28255468	28267904	12436
27	123088	123120	32	38	32469716	32489330	19614
28	143536	143672	136	39	84925632	84966030	40398
29	361190	361288	98	40	97443786	97508948	65162
30	418971	419205	234	41	256131058	256257452	126394

The class values are computed from Part I’s Theorem 1.1; for $n \leq 22$ they agree with the two direct enumerations made at intake. The first wrong odd term is $n = 23$. The entry’s definition, and its

links, are right. Barnabei et al. [5] cite Min’s recurrence for this sequence but print no value of it. This report records the correction; nothing has been submitted to the OEIS, and no erratum has been sent to the journal or to the author. OEIS data are available under CC BY-SA 4.0.

23.2 Further questions and research

1. **Effective constants and certified brackets.** (Part I, Section 9 and Section 11; Part II, Section 22.) The constants $D_{s,J}$, $H_{s,K}$ and the onsets of Theorem 9.1, and all remainder constants of Part II, are existence constants; no bracket is certified at a given input. *Possible route (the write’s suggestion, not carried out):* the singular structure is explicit (radicals, a single simple square-root singularity in $|x| < 1/2$), so the transfer remainders could be bounded on an explicit Delta contour. *Missing:* the explicit bounds and onsets. An uncertified indication of their size: the scaled errors of Proposition 18.2 computed at intake (note after it) suggest a constant near -450 in the $O(m^{-3})$ term of $w_{E,m}$.
2. **All-index monotonicity.** (Part I, Subsection 7.3: “an asymptotic conclusion, not a claimed all-index monotonicity proof”.) Is a_n nondecreasing for all n , and strictly increasing for $n \geq 5$? *Evidence (write, uncertified):* true for $n \leq 2001$ by the corrected recurrence. *Missing:* an injection $\mathcal{B}_n(I) \rightarrow \mathcal{B}_{n+1}(I)$, or explicit asymptotic error bounds (item 1) together with a finite check.
3. **The recurrence-defined sequence.** (Part I, Subsection 7.3: no claim concerning a natural boundary or non-D-finiteness of the old sequence.) What is the nature of $\hat{E}$, $\hat{O}$, defined by $\hat{E} = 1 + x\hat{O} + x^2\hat{E}\hat{O}(x^2)$, $\hat{O} = (1 - x\hat{E})^{-1}$ (Section 5)? Proposition 23.2 gives only $\hat{b}_n \leq a_n$, hence radius of convergence at least that of the class series. *Missing:* everything else; the question concerns the published sequence, not the class.
4. **Priority and Min’s thesis.** (Part I, abstract “Scope” and Section 11.) Min’s 2021 paper says its material is from her 2002 thesis, which neither manuscript nor the write inspected; bounded searches found no corrected enumeration (`data/149-enum-SOURCE_PROVENANCE.json`). *Missing:* the thesis, and an exhaustive search; no priority is claimed.
5. **Min–Park’s necessity lemmas inside the report.** (Part I, Subsection 2.1.) Part I accepts Theorem 4.1 and Corollaries 4.5 and 4.7 of Min–Park (2006) as published inputs; Ouchterlony’s Corollaries 4.2–4.3 and Lemma 4.4 give an independent route. *Missing:* a proof inside the report, which would remove the one external trust boundary of the enumeration; the companion’s exhaustive check for $n \leq 12$ is independent of the lemmas, its larger grammar check is not.
6. **Convergence, uniformity in the order, global expansions.** (Part I, Subsection 7.2 and Section 11; Part II, after Theorem 14.1.) Both Parts prove expansions at every fixed order only: no convergence of the formal series, no bound uniform in a growing order, no exponentially improved or global expansion, and in Part II no bound uniform in a growing window radius W .
7. **Fixed positive weight.** (Part II, “Scope” and Section 22.) The limit laws of Theorem 20.1 are for the sparse scaling $q = w/m$. A central limit theorem for the number of fixed-point pairs at fixed $q > 0$ (in particular $q = 1$, the uniform distribution on the class) is not asserted. *Possible route (the write’s suggestion, not carried out):* the bivariate system (13.5) is explicit, so the quasi-powers framework applies once the movement of the dominant singularity with q is controlled. *Missing:* that analysis.
8. **Inverses near the threshold and globally.** (Part II, Subsections 18.1–18.2, Section 22.) Theorem 19.1 holds on compact positive windows $a \leq w_0 \leq b$; Proposition 18.2 treats only the

two limiting targets at even m . Open: an inverse uniform as $a \downarrow 0$ or for targets approaching the threshold at an unspecified rate, the odd- m reverse behaviour beyond the obstruction (18.2), and a global complex inverse.

Not open. Min's Theorem 2.7, her list of values, and the data and program of A347546 are corrected above, with proofs. No question of either manuscript is answered by the other: Part II refines Part I's count and imports its decomposition; it does not touch the length asymptotics or the length inverses.

References

- [1] S. Min and S. Park, *The enumeration of doubly alternating Baxter permutations*, J. Korean Math. Soc. **43**(3) (2006), 553–561. [doi:10.4134/JKMS.2006.43.3.553](https://doi.org/10.4134/JKMS.2006.43.3.553). Theorem 4.1 and Corollaries 4.5, 4.7, printed pp. 558–560. (Part II's annotation: Theorem 4.1 and Corollaries 4.5, 4.7.)
- [2] S. Min, *The enumeration of involutions of doubly alternating Baxter permutations*, J. Chungcheong Math. Soc. **34**(3) (2021), 253–257. [doi:10.14403/jcms.2021.34.3.253](https://doi.org/10.14403/jcms.2021.34.3.253). Theorem 2.7 and its proof, especially printed p. 256. (Part II's annotation: Theorem 2.7 and its proof.)
- [3] OEIS Foundation Inc., A347546, <https://oeis.org/A347546>, source and table inspected 3 October 2026. The report distinguishes the intended class from the then-tabulated recurrence sequence. (Part II's annotation: The intended class and the printed recurrence are distinguished in Report 149; the associated source review was dated 3 October 2026.) Read again by the write on 5 October 2026: revision #20 of June 29, 2022, data unchanged.
- [4] E. Ouchterlony, *Pattern avoiding doubly alternating permutations*, FPSAC 2006 proceedings, Section 4. <https://garsia.math.yorku.ca/fpsac06/papers/83.pdf>.
- [5] M. Barnabei, F. Bonetti, N. Castronuovo, and M. Silimbani, *Pattern Avoiding Alternating Involutions*, Enumerative Combinatorics and Applications **3**:1 (2023), Article S2R4, Section 10. https://ecajournal.haifa.ac.il/Volume2023/ECA2023_S2A4.pdf. arXiv:2206.13877.
- [6] R. Brignall, S. Huczynska, and V. Vatter, *Simple permutations and algebraic generating functions*, J. Combin. Theory Ser. A **115**(3) (2008), 423–441. [doi:10.1016/j.jcta.2007.06.007](https://doi.org/10.1016/j.jcta.2007.06.007). Corollary 1.4 and Theorem 5.3 in the author-hosted version (author manuscript), <https://www.rbrignall.org.uk/papers/simple-enum.pdf>.
- [7] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, Chapter VI. <https://algo.inria.fr/flajolet/Publications/book.pdf>.
- [8] *Corrected enumeration and asymptotics of alternating Baxter involutions*, Report 149, 3 October 2026. Source and PDF included unchanged in the accompanying foundation directory. Sections 2–4 supply the precise structural dependency and all-length decomposition. [write] Printed as Part I of this report, with the same section numbers; the foundation copy is not shipped.
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